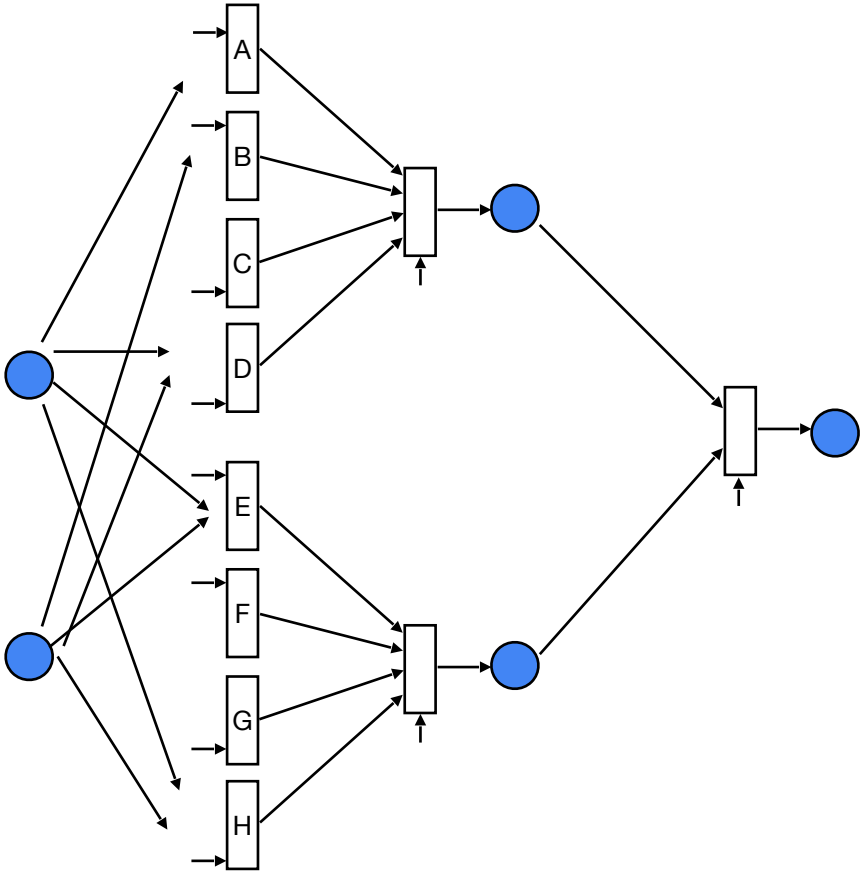
Introduction to Deep Learning

CS/EE/CMS 148a - Winter 2026 - P. Perona

Lecture 9 - RNNs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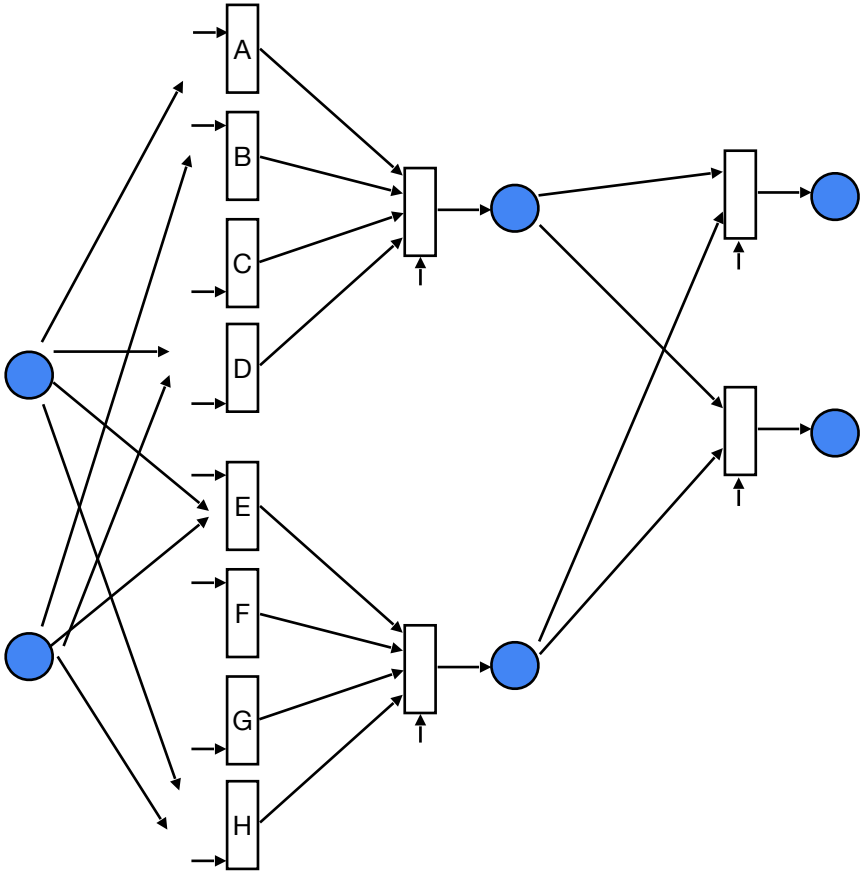
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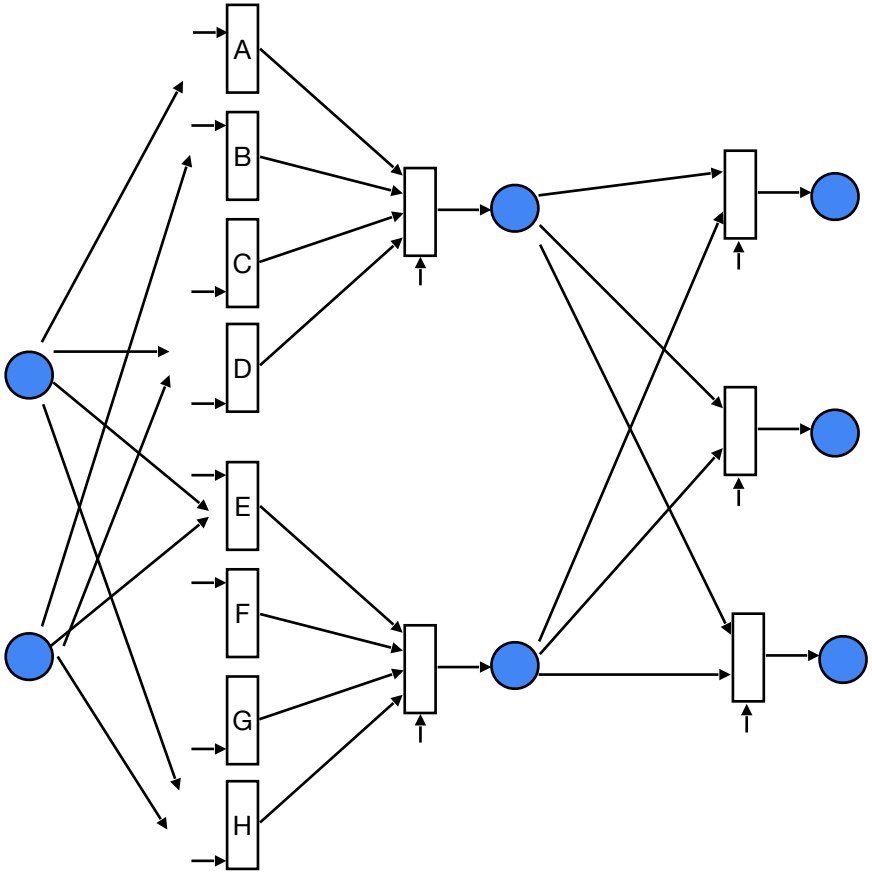
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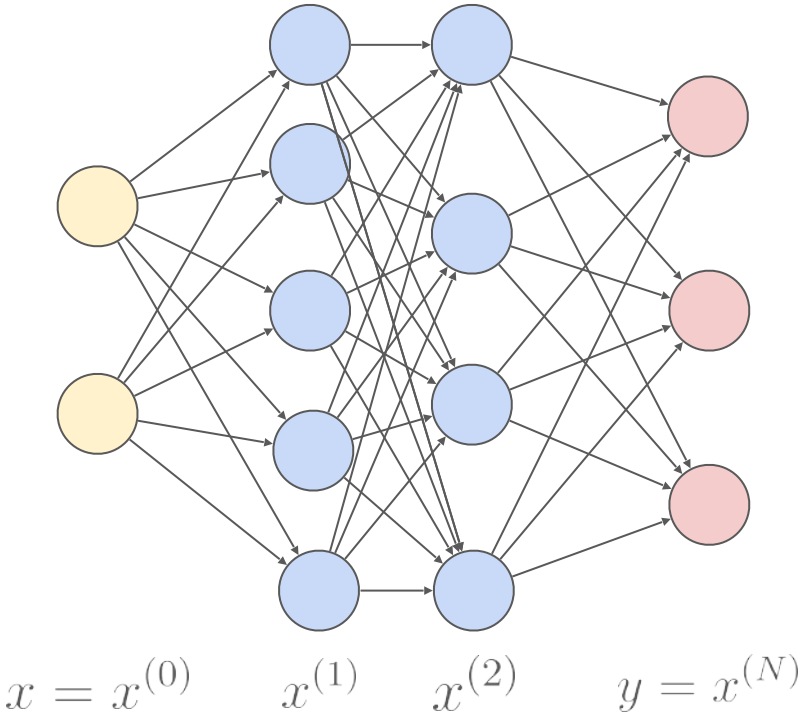
Deep networks: tasks so far



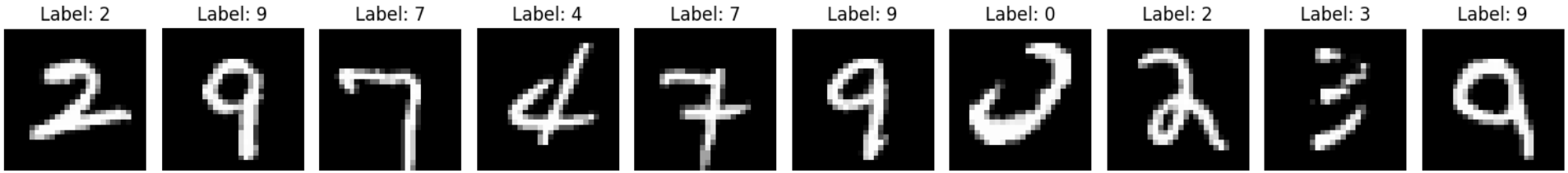
Classify an animal based on height and weight

Classify a handwritten digit

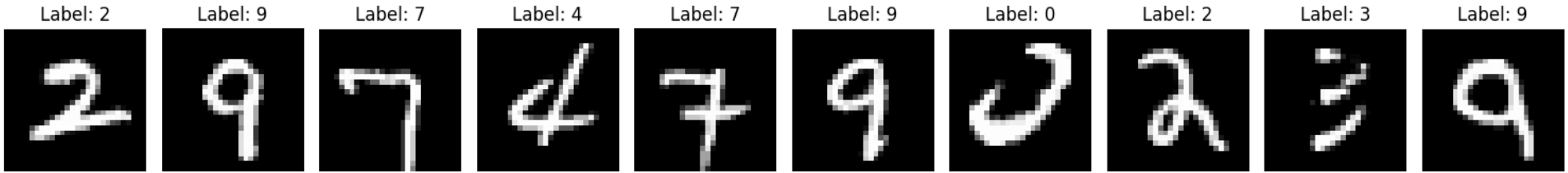
Predict the price of a house given distance from public transportation, age, size,...

Detect if there is a tiger in an image

The data samples are independent

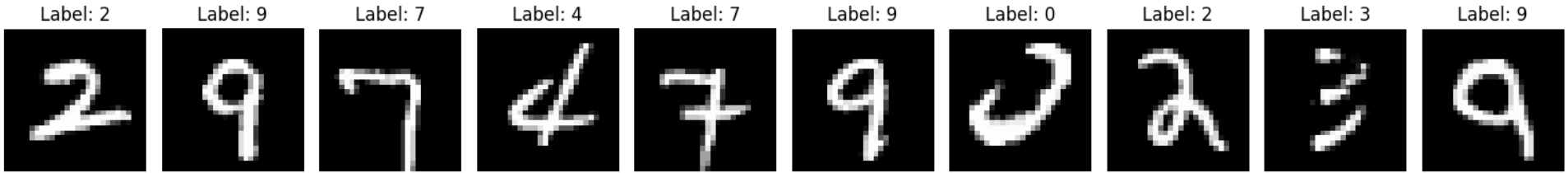


The data samples are independent



$$\mathcal{D} = \{x_i, y_i\}$$

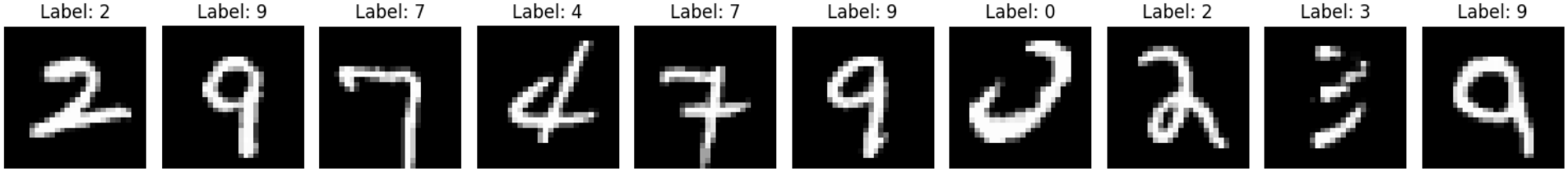
The data samples are independent



$$\mathcal{D} = \{x_i, y_i\}$$

$$f : X \longrightarrow Y$$

The data samples are independent



$$\mathcal{D} = \{x_i, y_i\}$$

Observed independently

$$f : X \longrightarrow Y$$

Independence:

Independence:

$$P(y_1, y_2, \ldots, y_N \mid x_1, x_2, \ldots, x_N) = \prod_{i=1}^{N} P(y_i \mid x_1, \ldots, x_N) = \prod_{i=1}^{N} P(y_i \mid x_i)$$

Independence:

$$P(y_1, y_2, \ldots, y_N \mid x_1, x_2, \ldots, x_N) = \prod_{i=1}^{N} P(y_i \mid x_1, \ldots, x_N) = \prod_{i=1}^{N} P(y_i \mid x_i)$$

$$\log P(y \mid x) = \sum_{i=1}^{N} \log P(y_i \mid x_i)$$

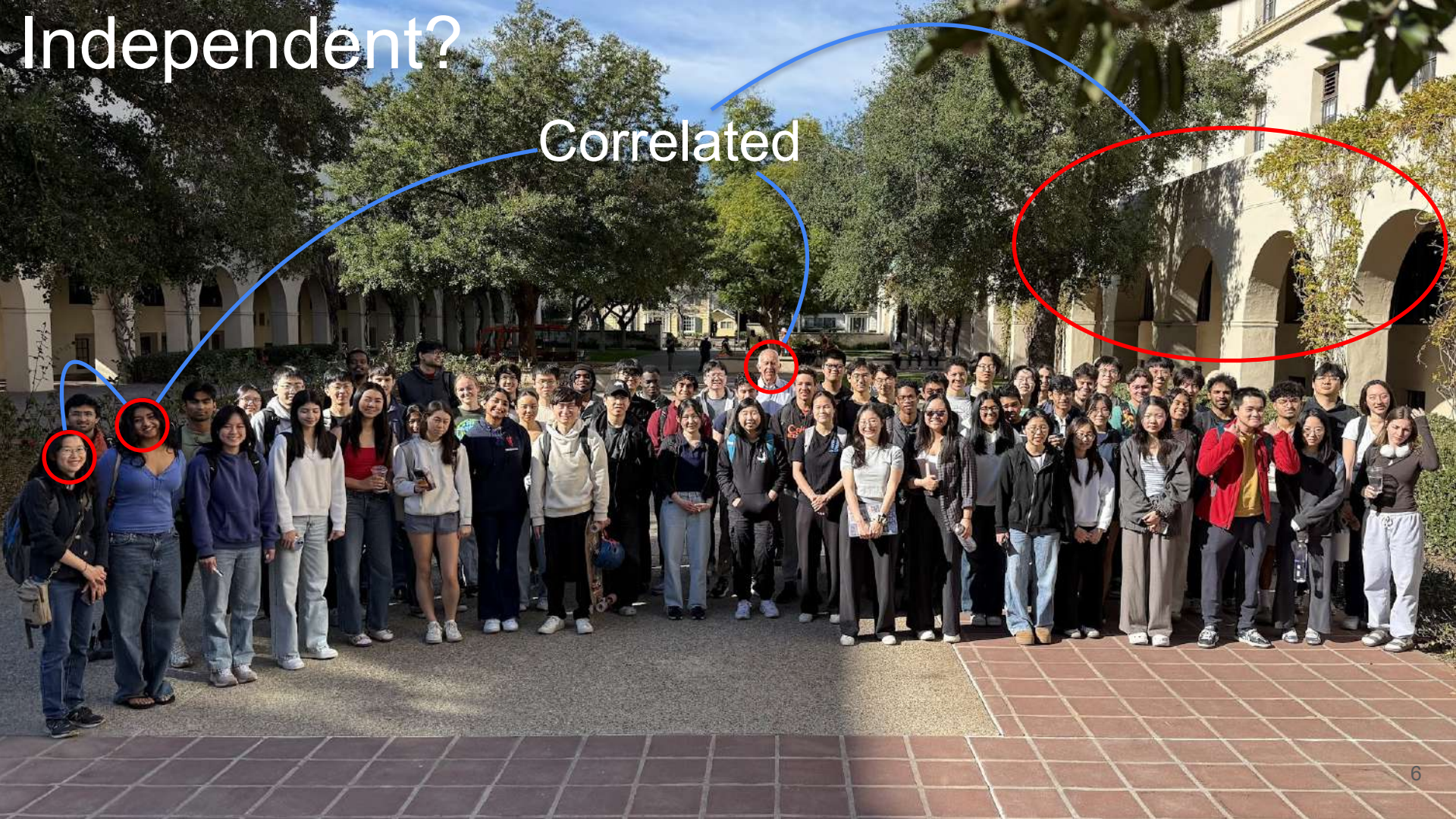
Independence:

$$P(y_1, y_2, \ldots, y_N \mid x_1, x_2, \ldots, x_N) = \prod_{i=1}^{N} P(y_i \mid x_1, \ldots, x_N) = \prod_{i=1}^{N} P(y_i \mid x_i)$$

$$\log P(y \mid x) = \sum_{i=1}^{N} \log P(y_i \mid x_i)$$

$$\mathscr{L}(w) = \sum_{i=1}^{N} \mathscr{L}'(f(x_i, w), y_i)$$

Independent?


Independent?
Correlated

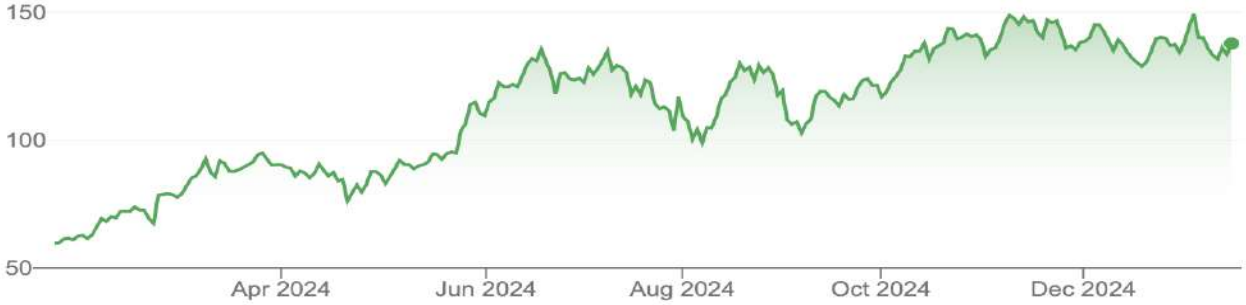
Sequence data

Words in text

We hold these truths to be self-evident, that all men are created equal, that they are endowed by their Creator with certain unalienable Rights, that among these are Life, Liberty and the pursuit of Happiness.

Frames in video

Stock price over time



Tasks

Tasks

- Classification

Tasks

- Classification
  - Sentiment analysis

Tasks

- Classification
  - Sentiment analysis
  - Activity/action recognition

Tasks

- Classification
  - Sentiment analysis
  - Activity/action recognition
- Prediction

Tasks

- Classification
  - Sentiment analysis
  - Activity/action recognition
- Prediction
  - Next day stock price

Tasks

- Classification
  - Sentiment analysis
  - Activity/action recognition
- Prediction
  - Next day stock price
- Sequence to sequence

Tasks

- Classification
  - Sentiment analysis
  - Activity/action recognition
- Prediction
  - Next day stock price
- Sequence to sequence
  - Language translation

Tasks

- Classification
  - Sentiment analysis
  - Activity/action recognition
- Prediction
  - Next day stock price
- Sequence to sequence
  - Language translation
  - Music transcription

Tasks

- Classification
  - Sentiment analysis
  - Activity/action recognition
- Prediction
  - Next day stock price
- Sequence to sequence
  - Language translation
  - Music transcription
  - Speech recognition

Tasks

- Classification
  - Sentiment analysis
  - Activity/action recognition
- Prediction
  - Next day stock price
- Sequence to sequence
  - Language translation
  - Music transcription
  - Speech recognition
- Generation

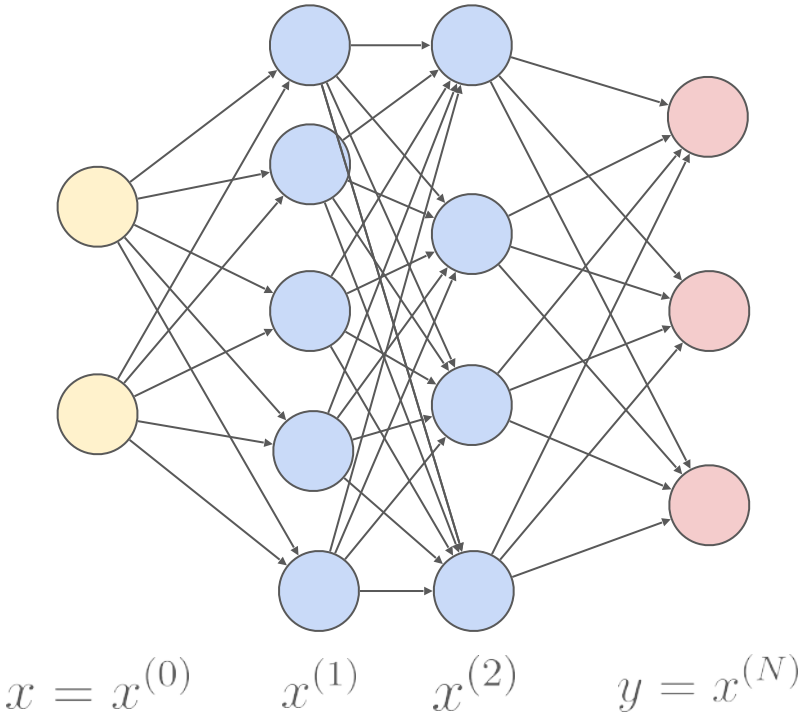
Tasks

- Classification
  - Sentiment analysis
  - Activity/action recognition
- Prediction
  - Next day stock price
- Sequence to sequence
  - Language translation
  - Music transcription
  - Speech recognition
- Generation
  - Summarization

Tasks

- Classification
  - Sentiment analysis
  - Activity/action recognition
- Prediction
  - Next day stock price
- Sequence to sequence
  - Language translation
  - Music transcription
  - Speech recognition
- Generation
  - Summarization
  - Question answering and conversation

Autoregression


$x = x^{(0)}$
$x^{(1)}$
$x^{(2)}$
$y = x^{(N)}$

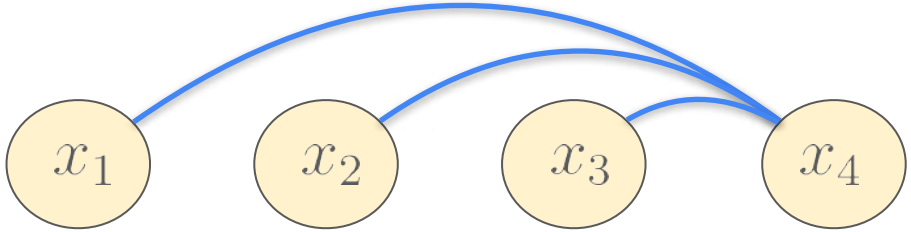
$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2)p(x_3)p(x_4)$$

Moving?

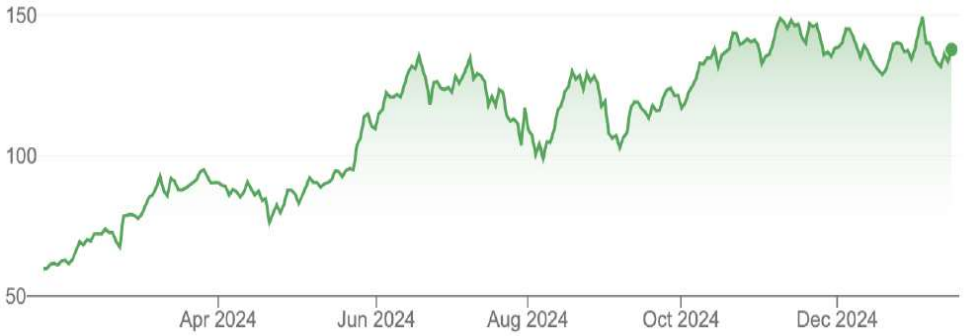
x_1 x_2 x_3 x_4

$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2)p(x_3)p(x_4)$$

Moving?



$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3)$$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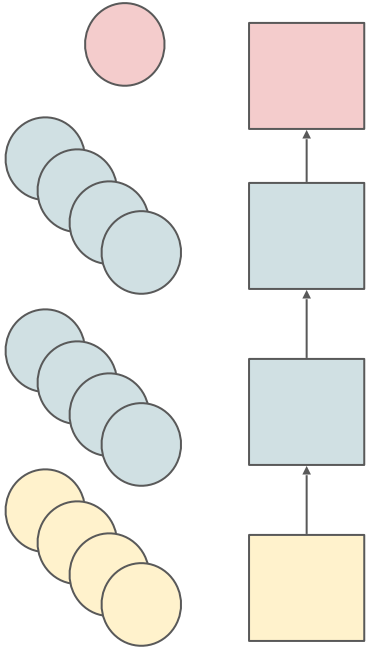

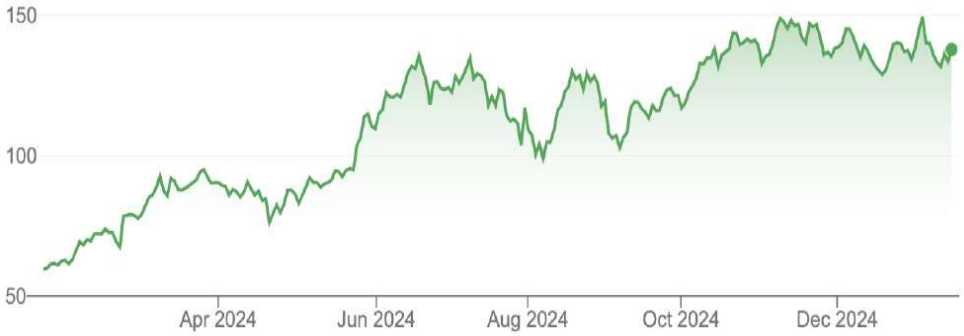
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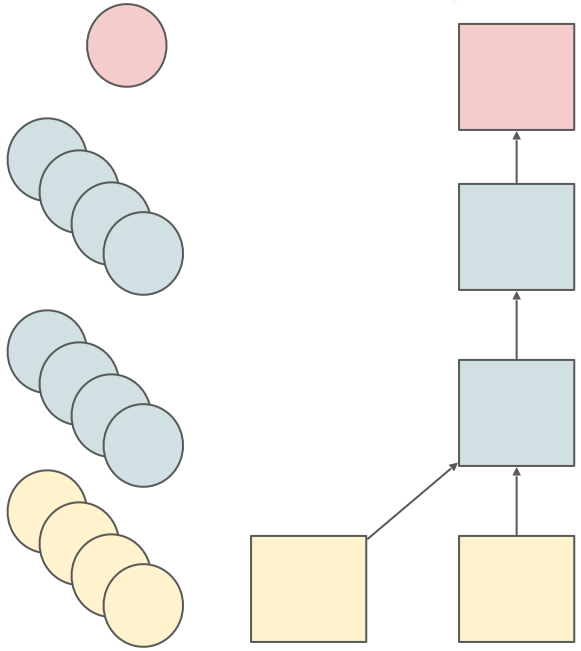

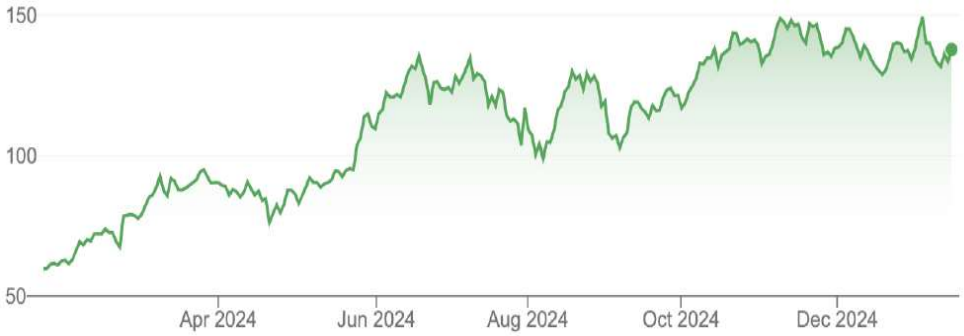
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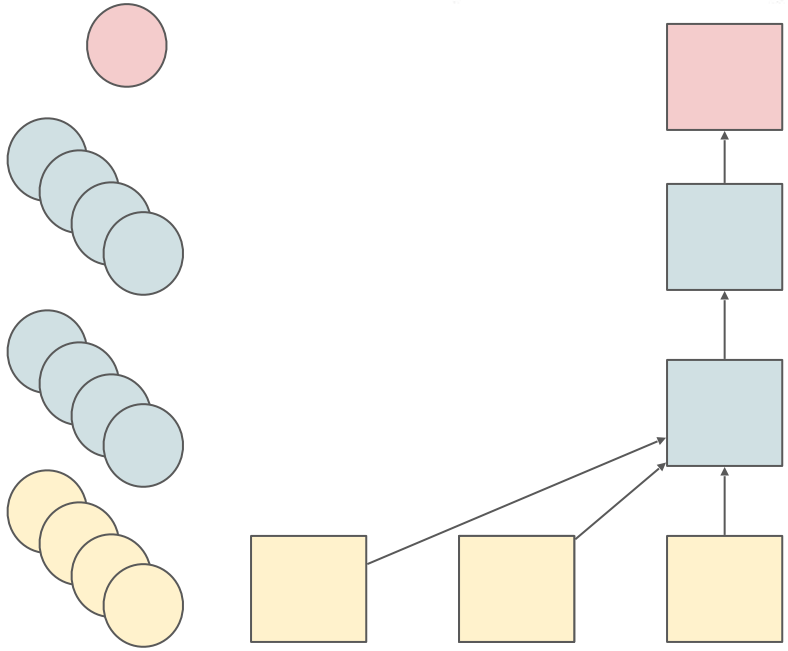

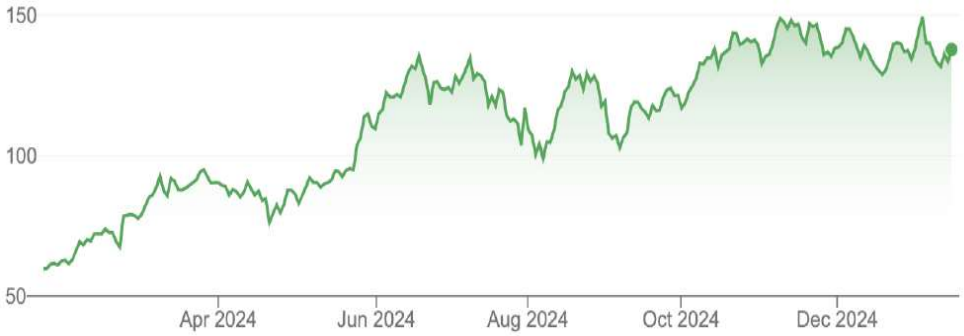
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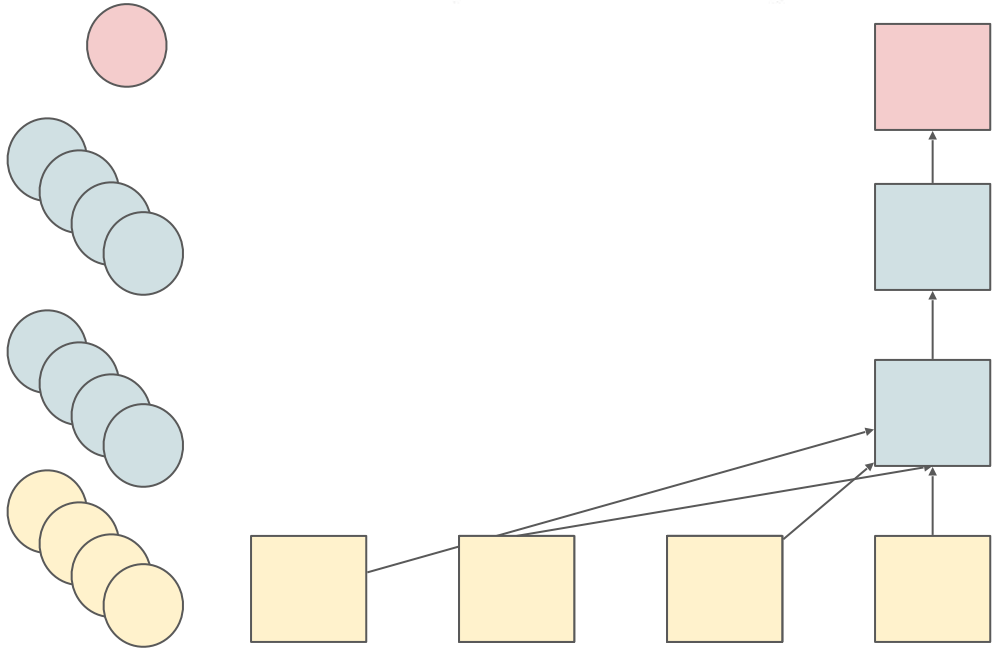

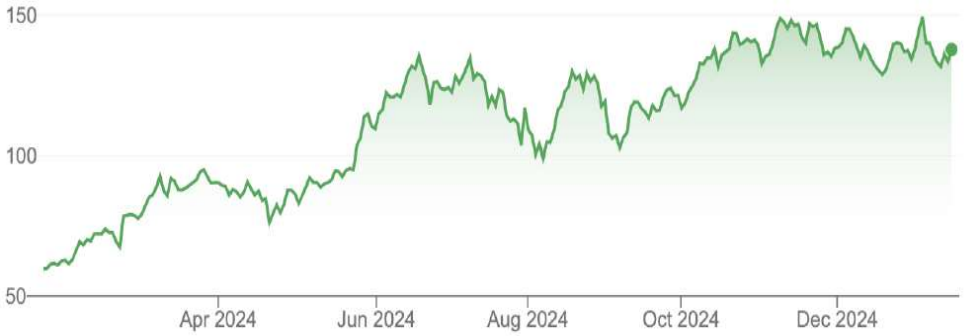
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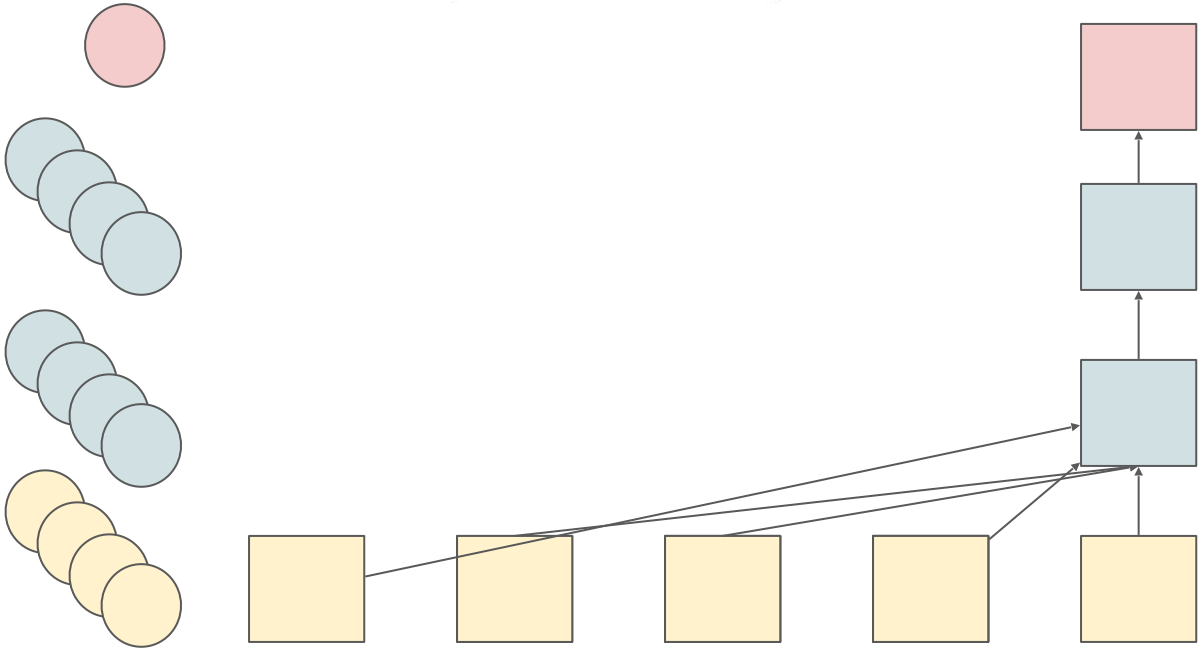

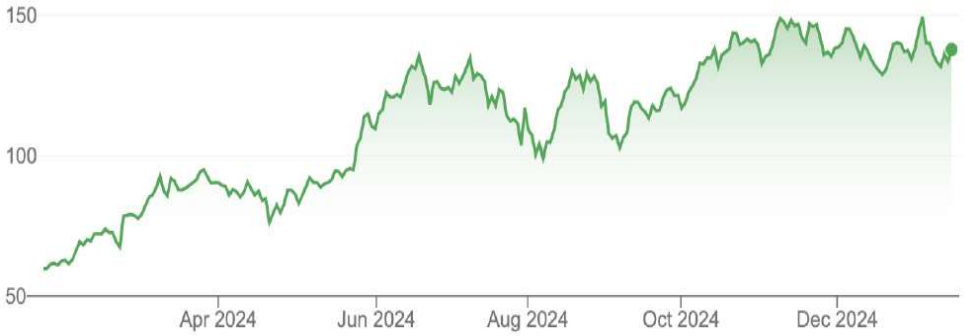
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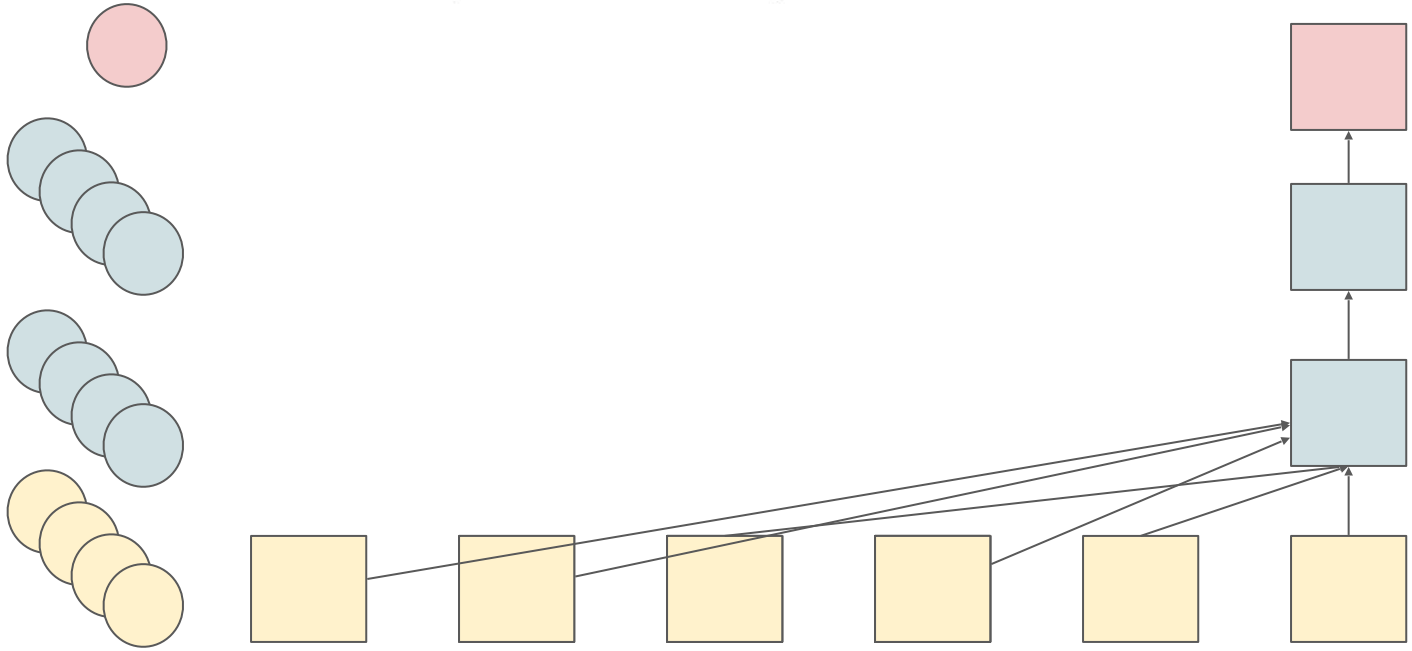
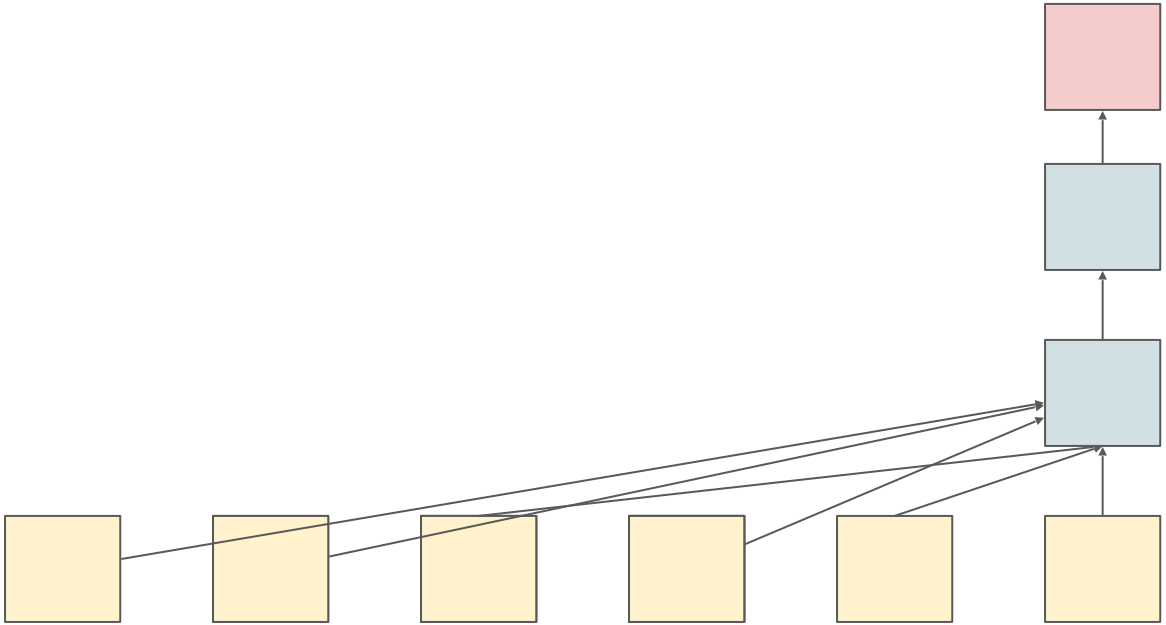
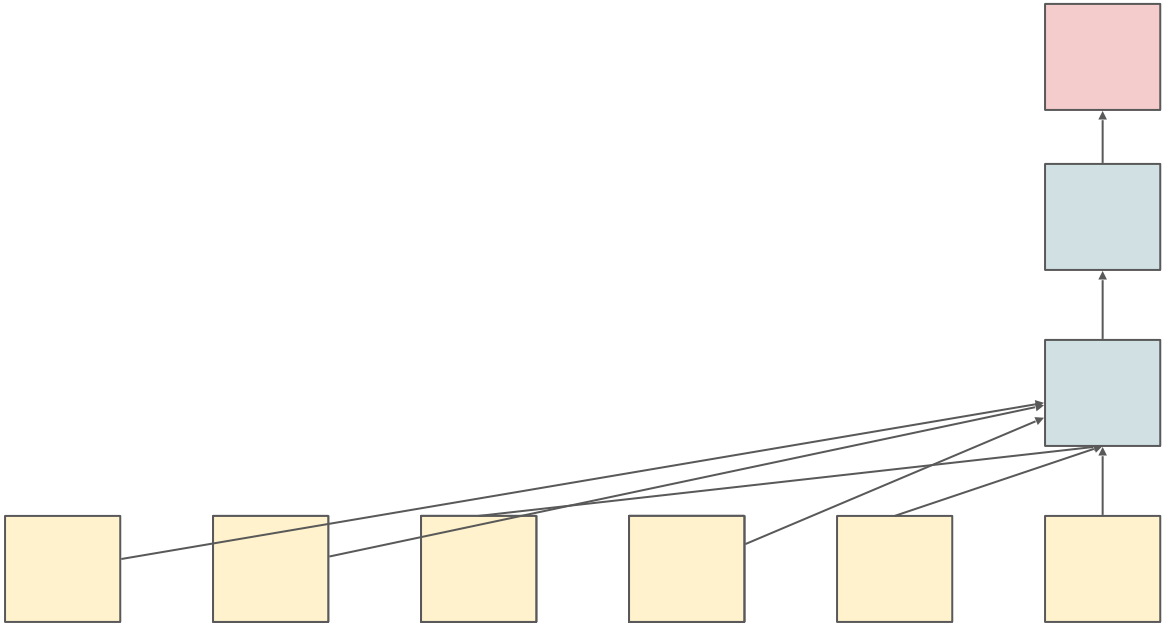


Output depends directly on T steps



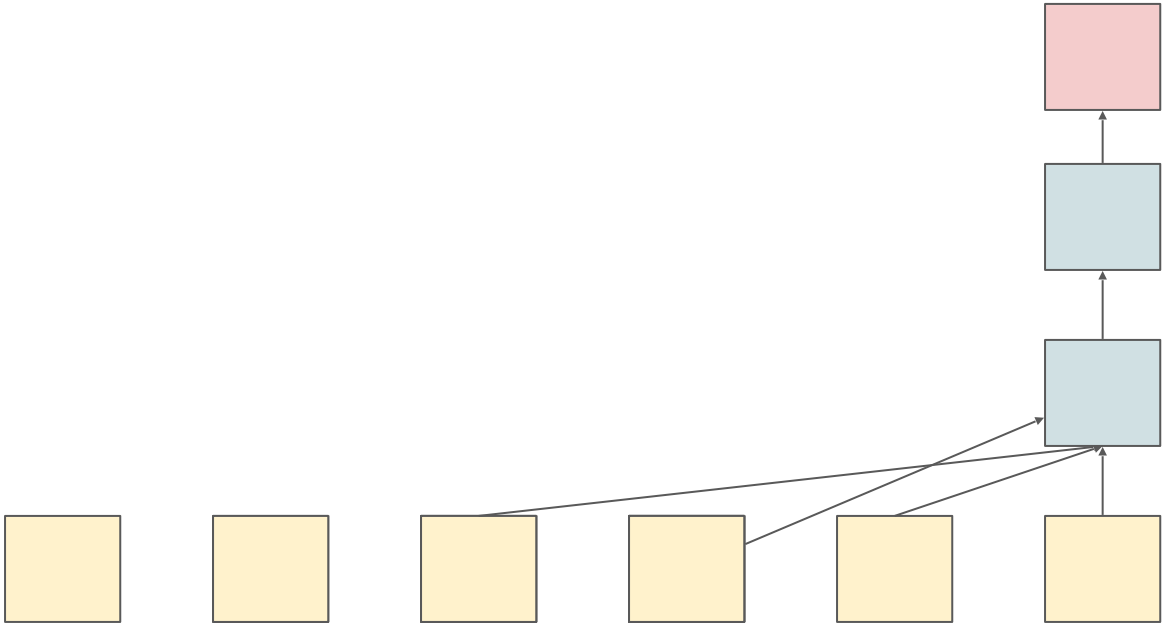
Output depends directly on T steps

What happens if T gets very large?

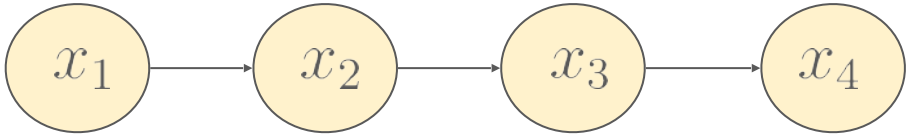


What happens if T gets very large?

Learning can get hard!
Gradients can explode or vanish very easily



Markov assumption

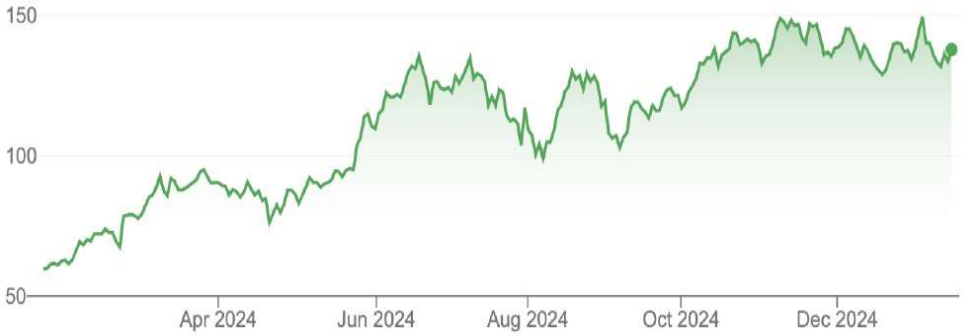


$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4|x_3)$$

Second order Markov assumption

$$x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_4$$

$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2|x_1)p(x_3|x_2, x_1)p(x_4|x_3, x_2)$$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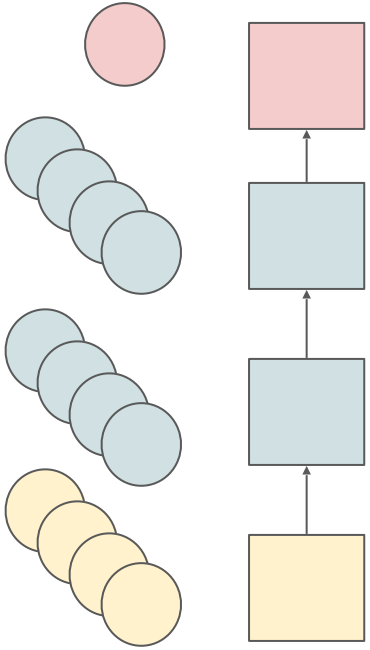

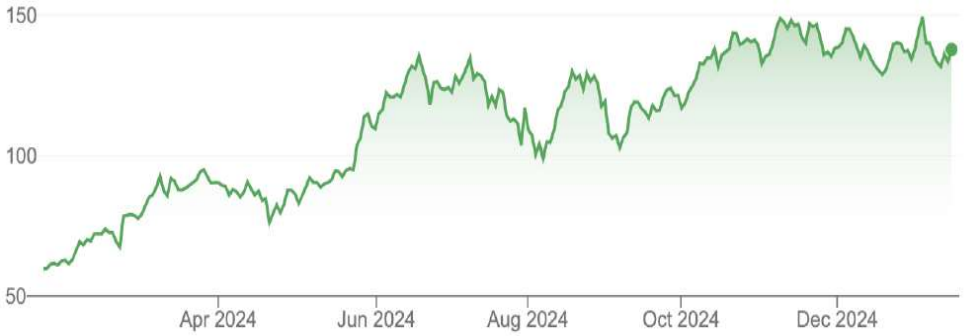
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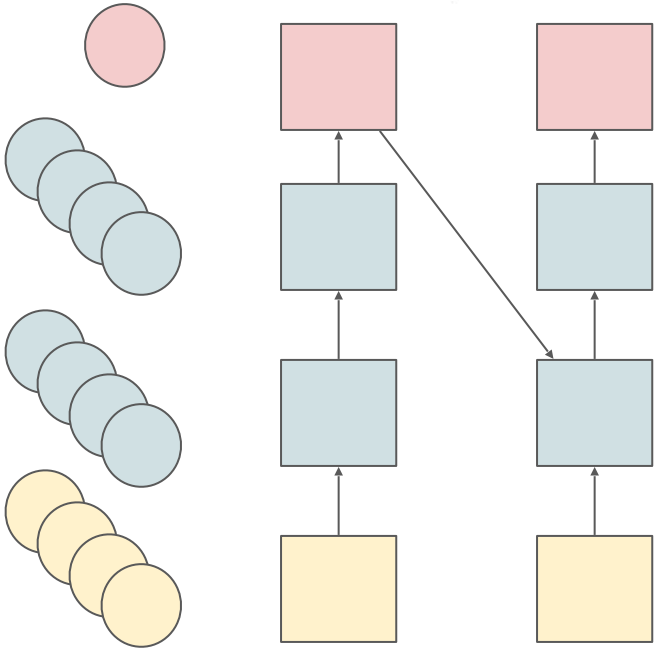
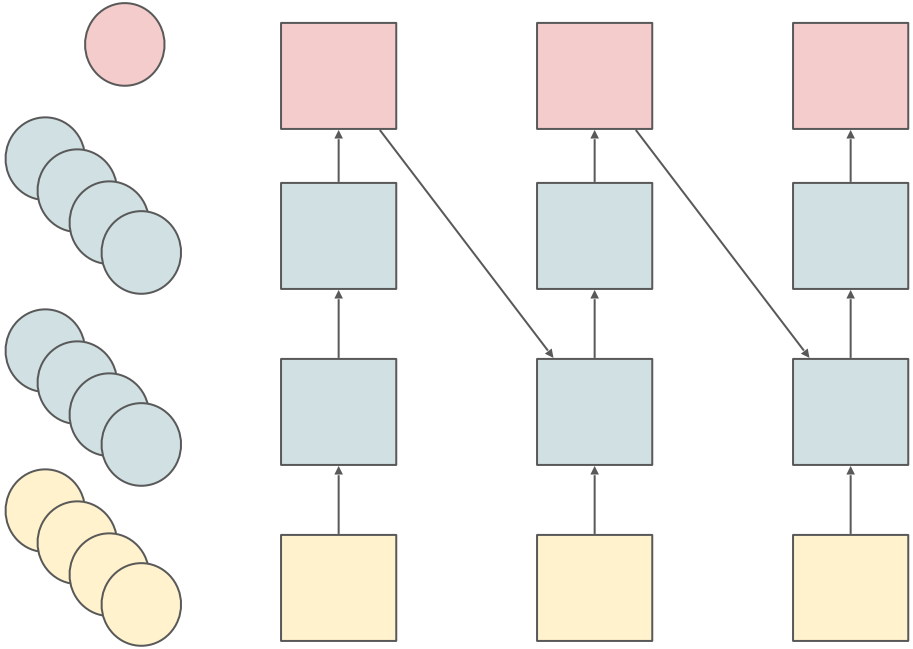
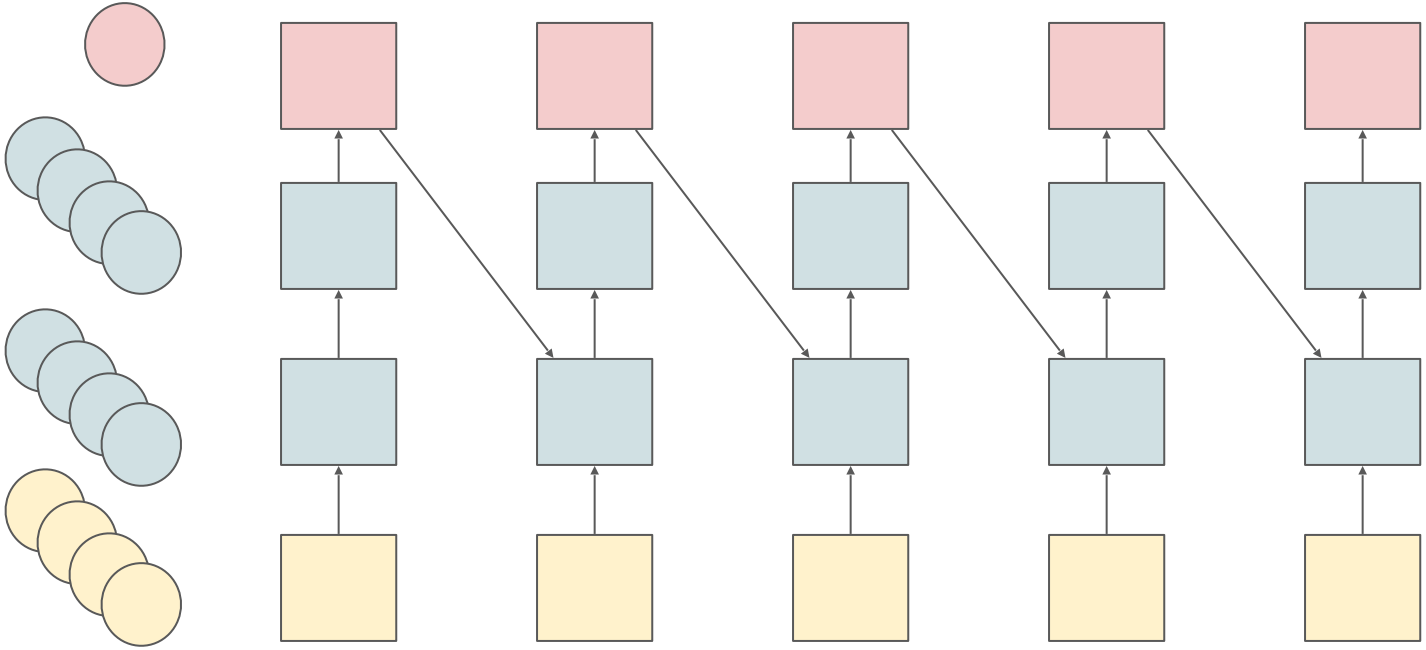


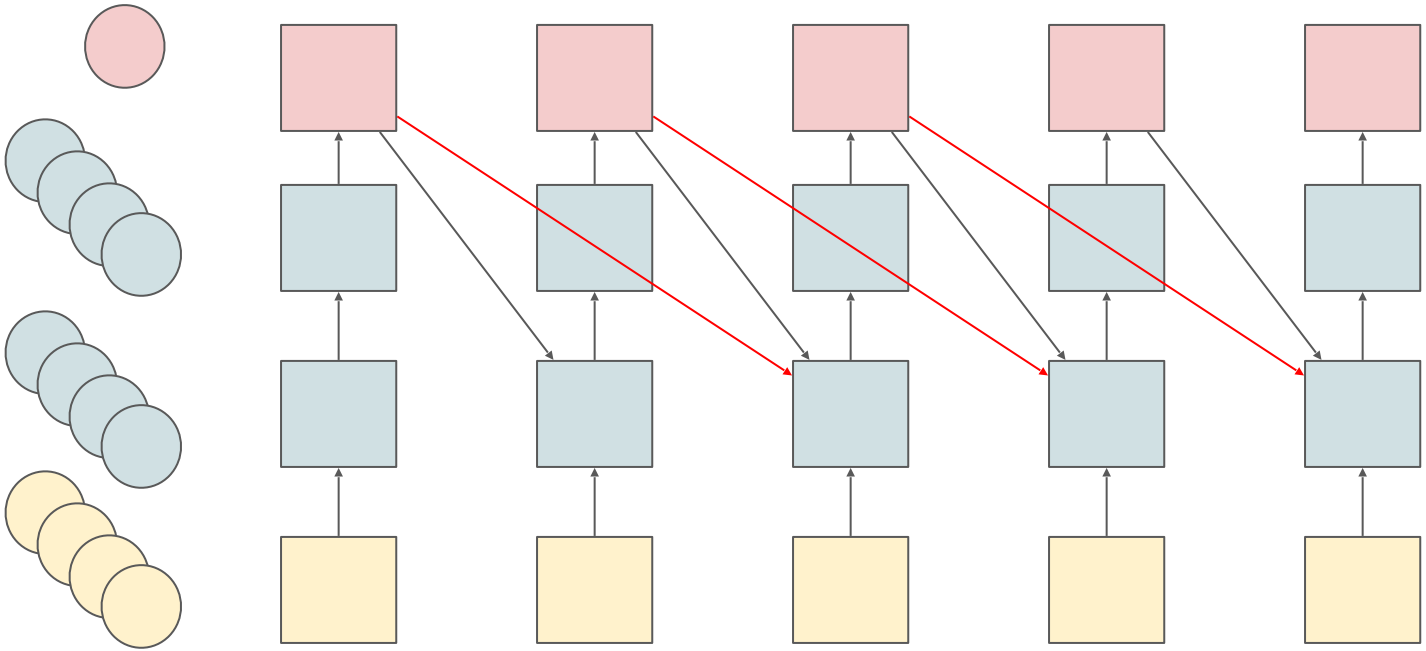
NARX - Non-linear autoregressive network with exogenous input



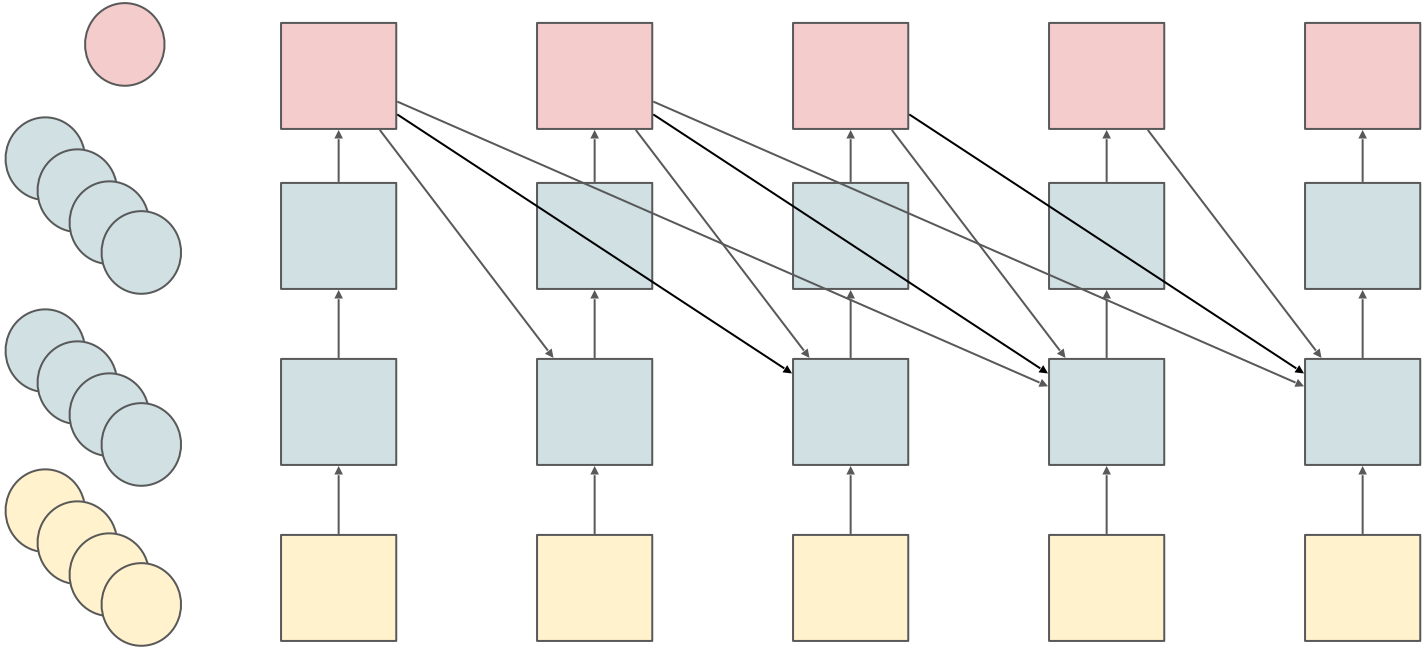
NARX



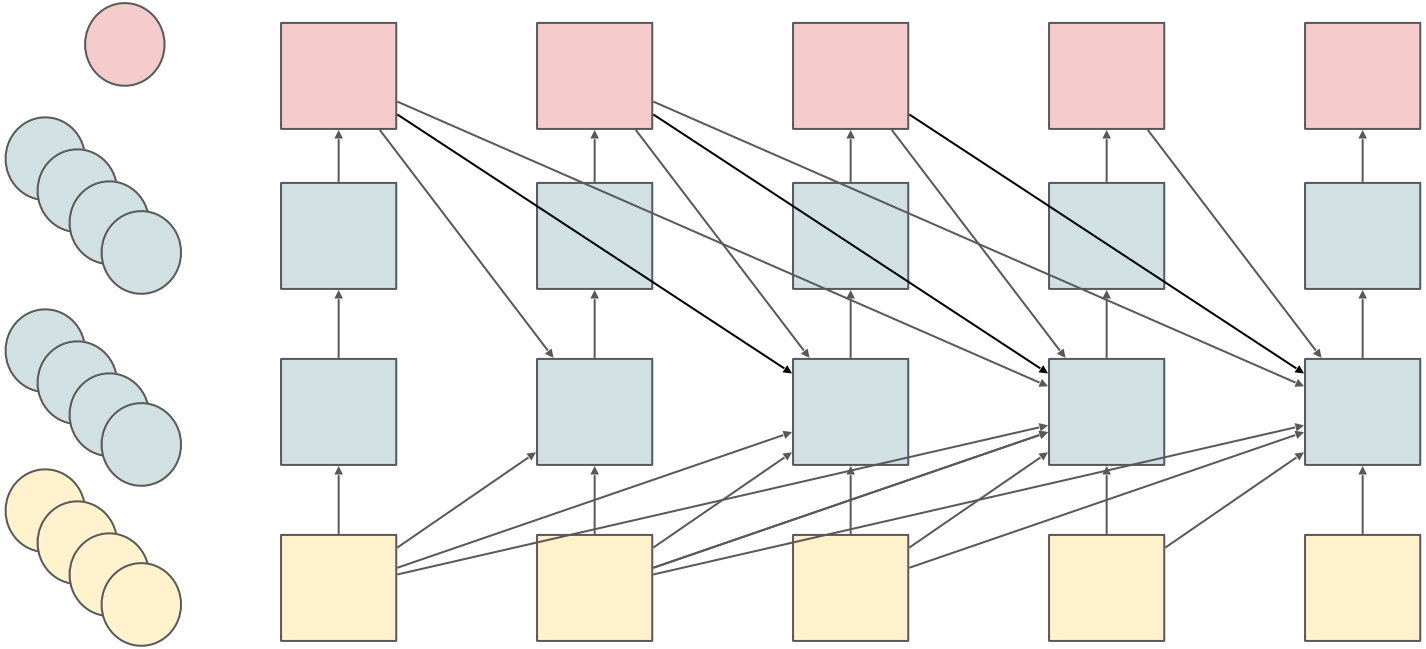
NARX



NARX

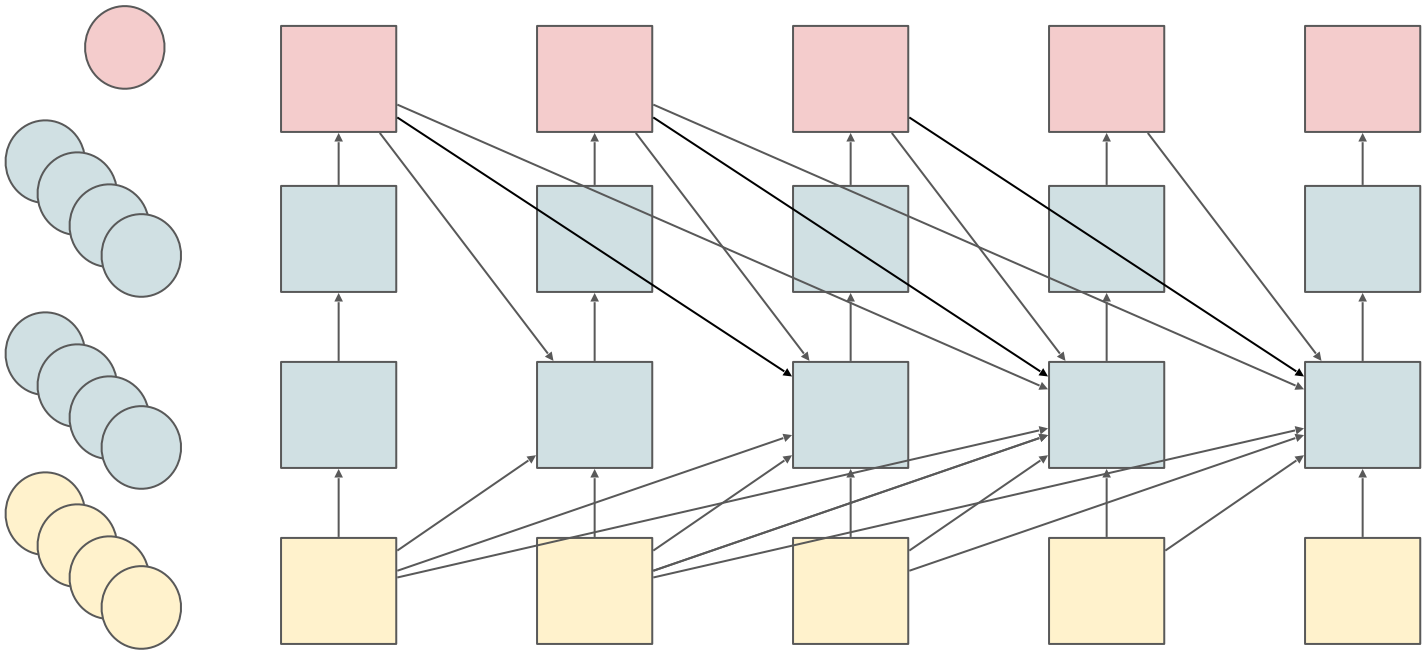


NARX



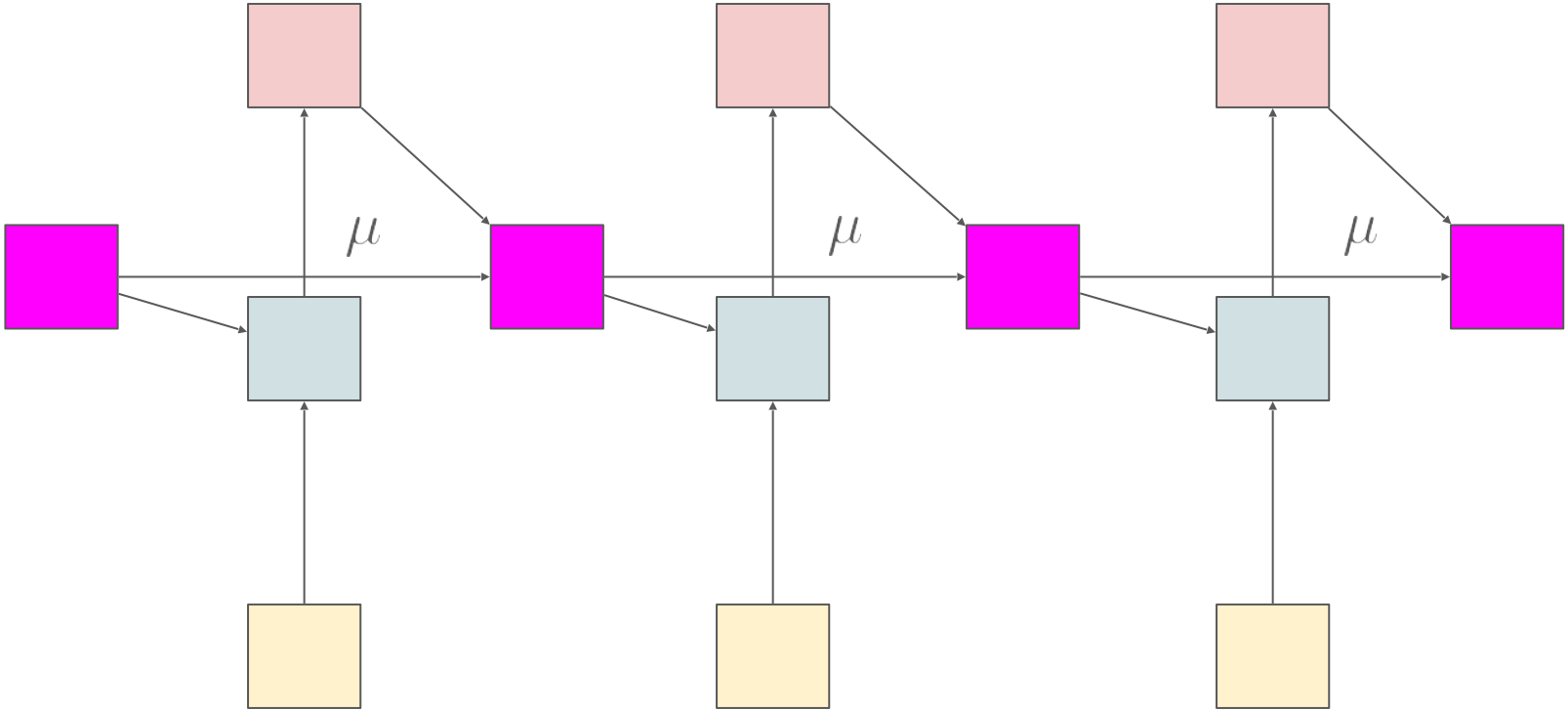
NARX

Output also has a “memory” role….

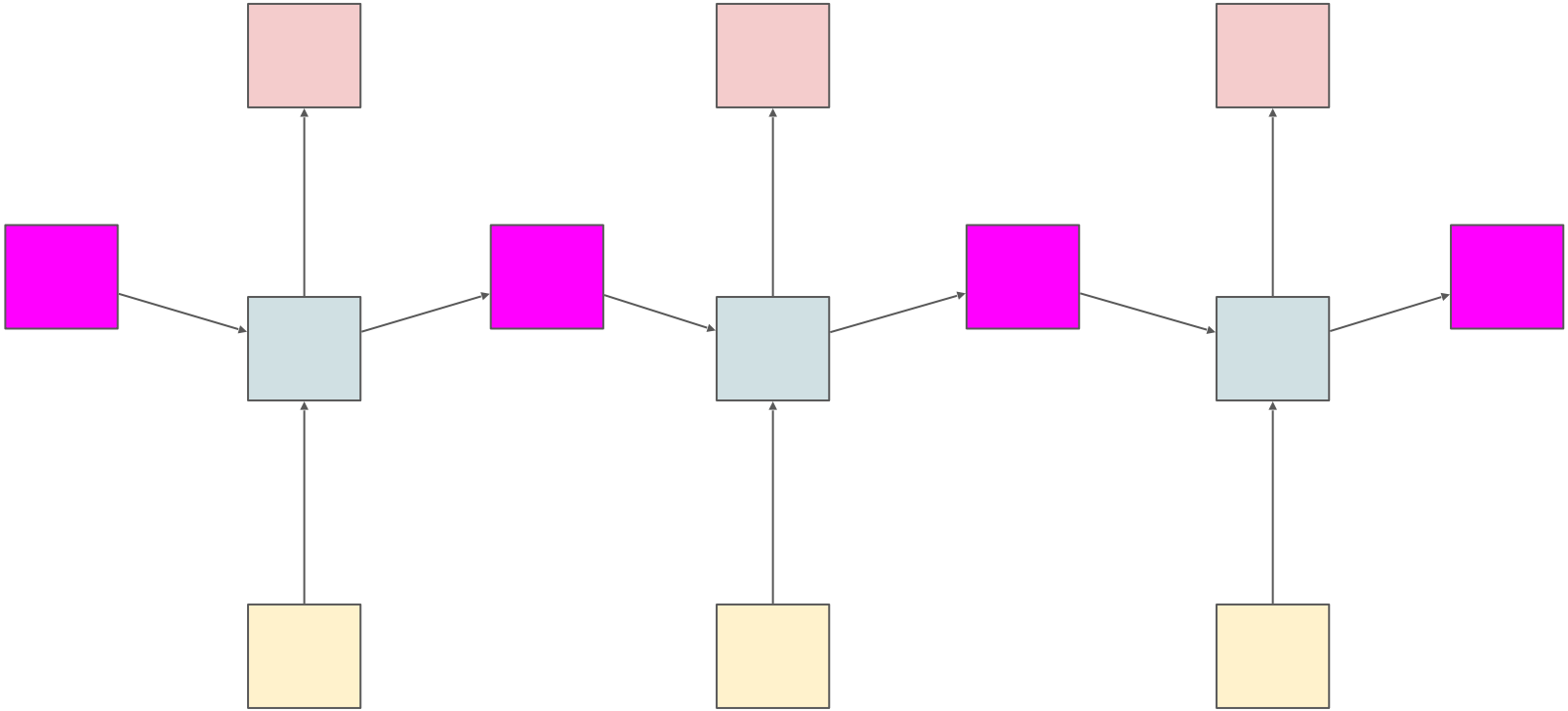


Recurrent Neural Networks

Jordan Networks (1986)

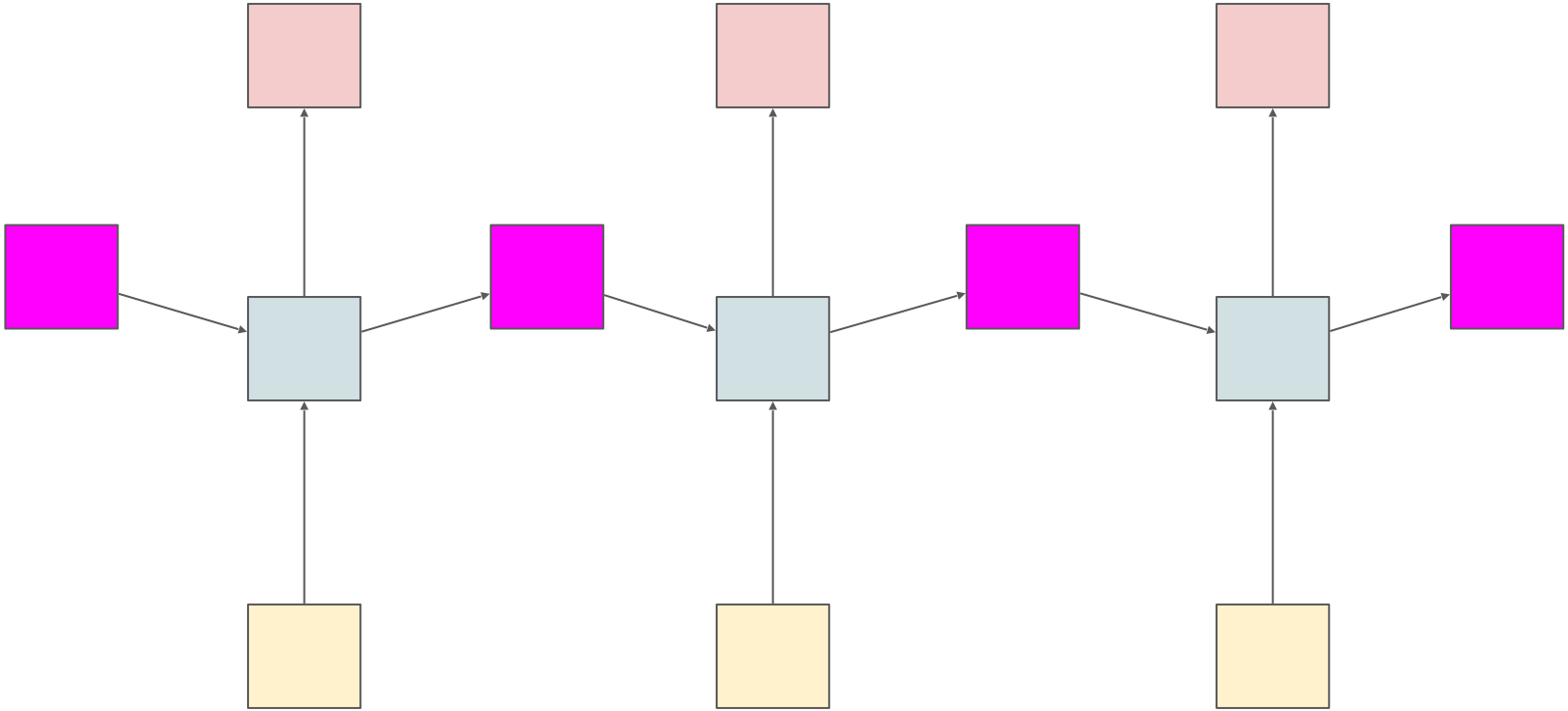


Elman Networks (1990)

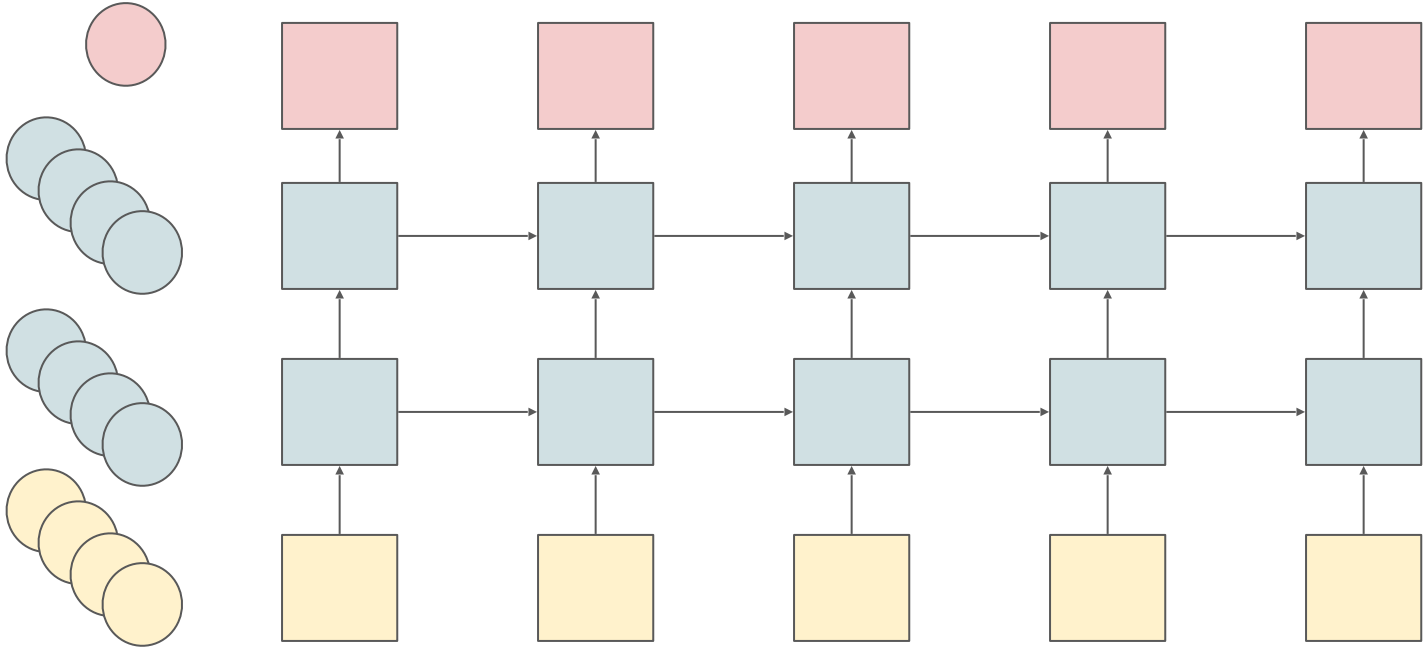


Elman Networks (1990)

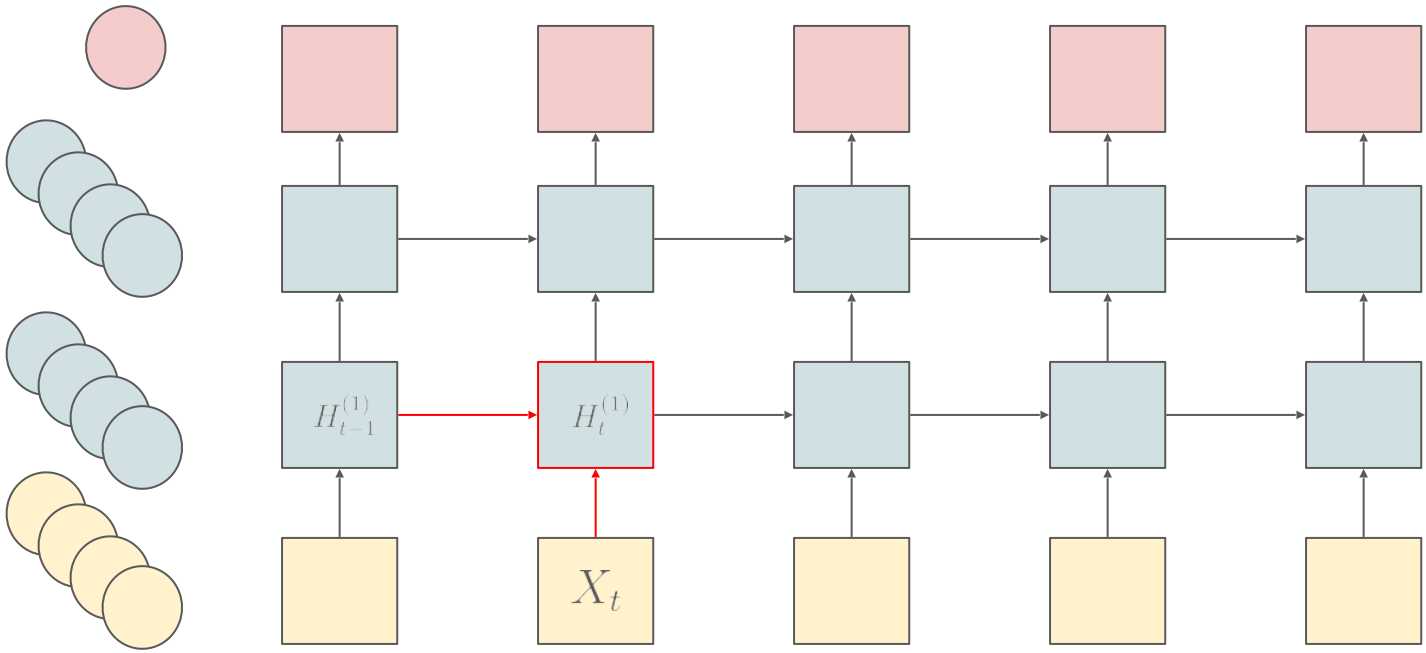
“Memory” is a clone of the hidden state.
The input weight into the next hidden state is learned.



RNN - Recurrent Neural Networks (1990s)



RNN - Recurrent Neural Networks



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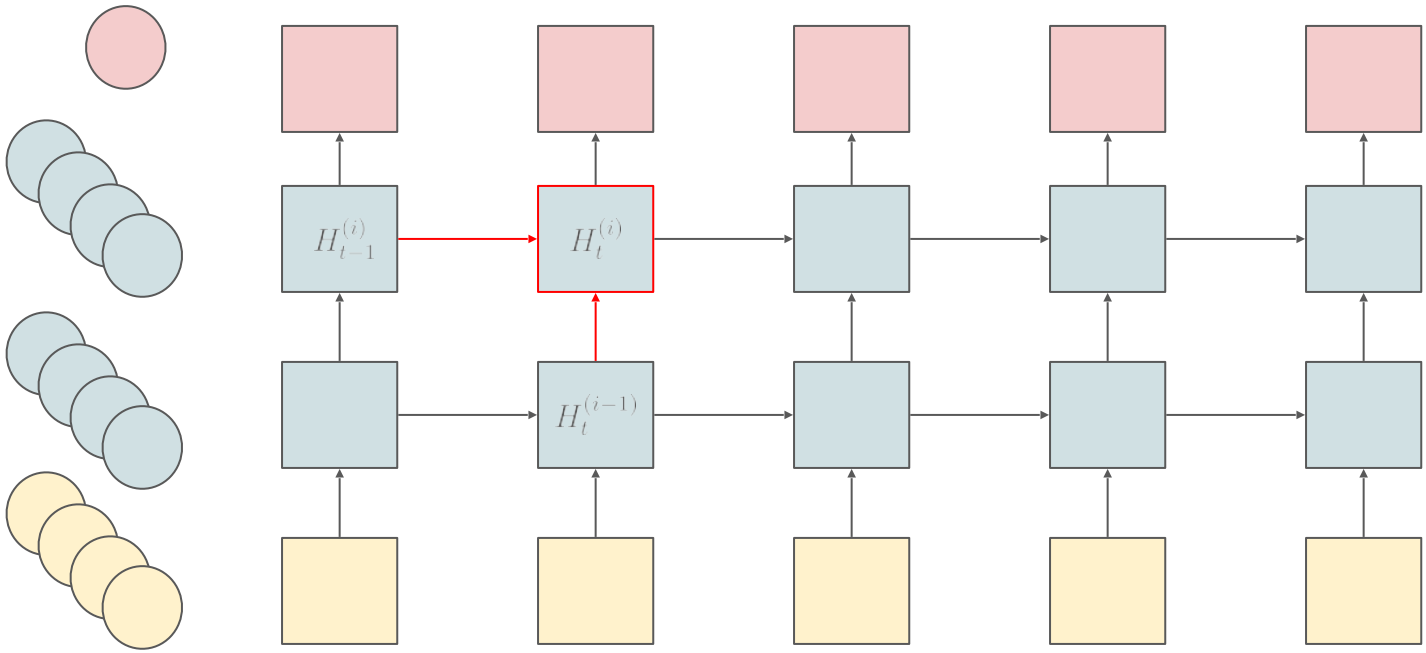

$$H_t^{(i)} = \sigma([H_{t-1}^{(i)} H_t^{(i-1)}] W^{(i,i-1)})$$

Language Models

Language models

We hold these truths to be self-evident, that all men are created equal, that they are endowed by their Creator with certain unalienable Rights, that among these are Life, Liberty and the pursuit of Happiness.

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Model the probability distribution of words (or token) sequences

Can be used to

- assess the likelihood of a sentence
- predict the next word in a sentence
- generate sentences

Language models and Markov assumption

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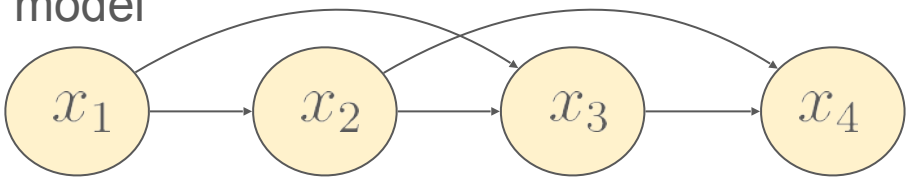
Bigram language model



$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2|x_1)p(x_3|x_2, x_1)p(x_4|x_3, x_2)$$

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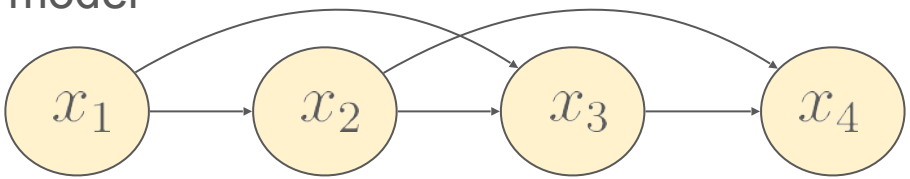

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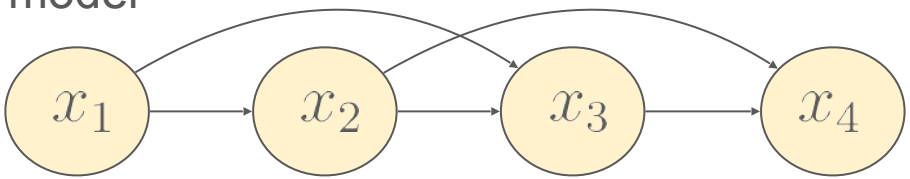

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P(“we”) P(“hold” | “we”)

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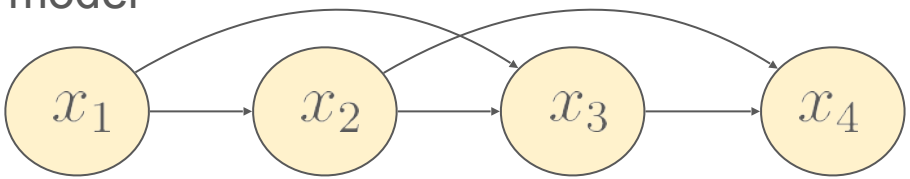

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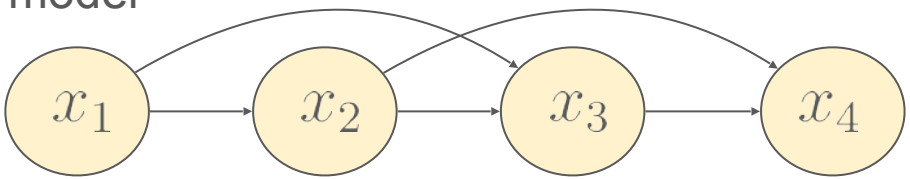

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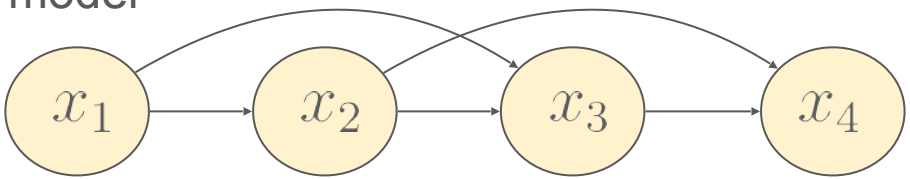

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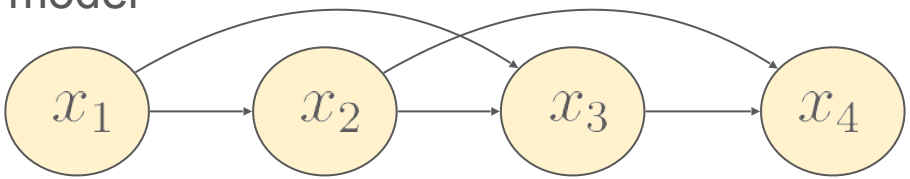

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…

What about rare words combinations?

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$$p(z \mid x, y) = \frac{N(x, y, z)}{N(x, y)}$$

P(heads)?

P(heads)?

P(heads)?

LIBERTY
IN GOD WE TRUST
2022
P
LIBERTY
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2022
P
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2022
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P(heads)?

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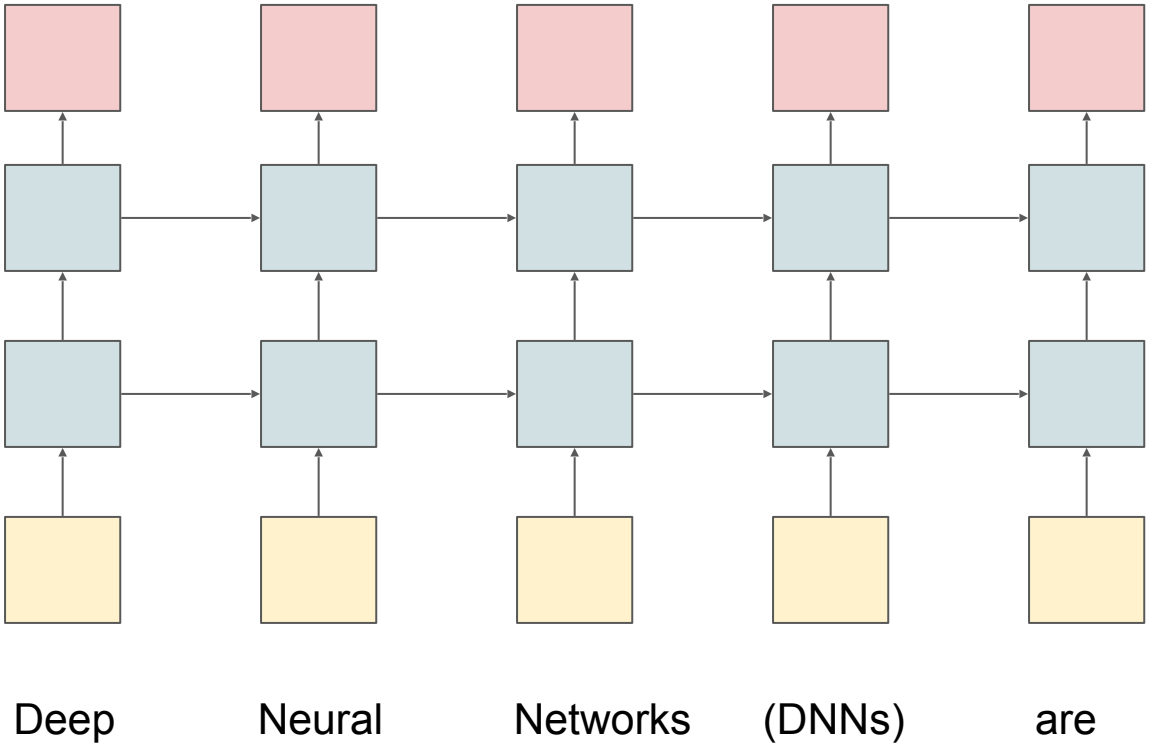
$$p(z \mid x, y) = \frac{N(x, y, z)}{N(x, y)}$$

Regularization by Laplace smoothing

$$p(z \mid x, y) = \frac{N(x, y, z) + \alpha p(z)}{N(x, y) + \alpha}$$


Deep
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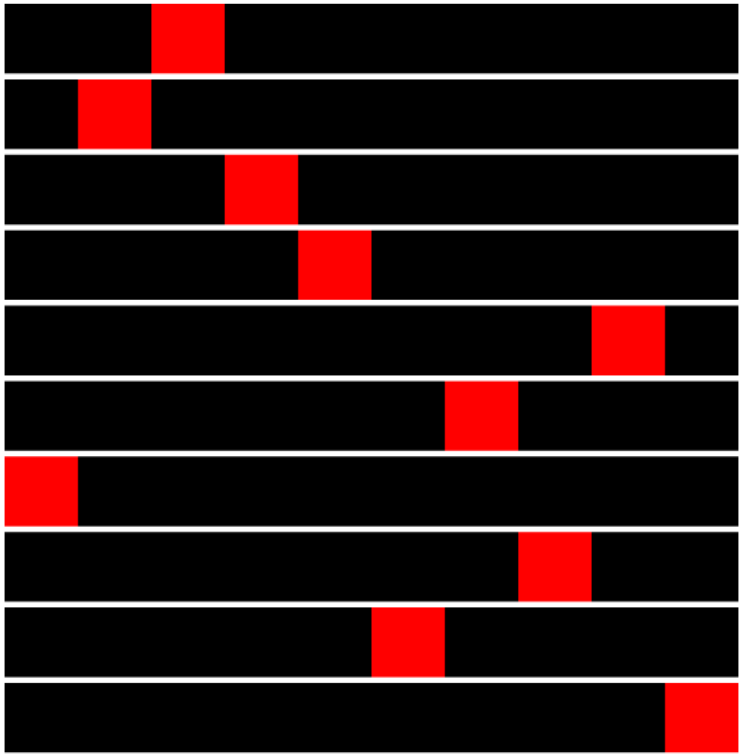

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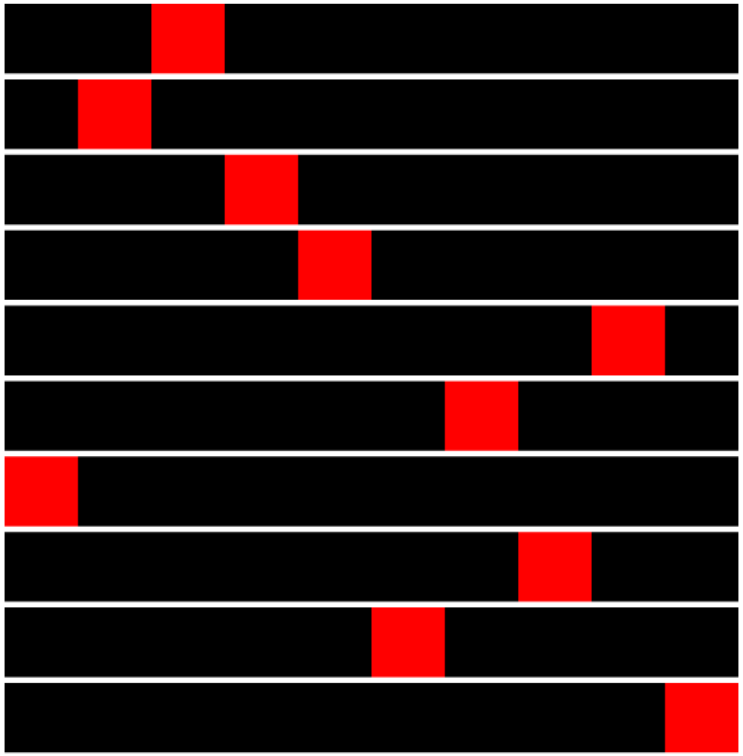


Maps to real numbers

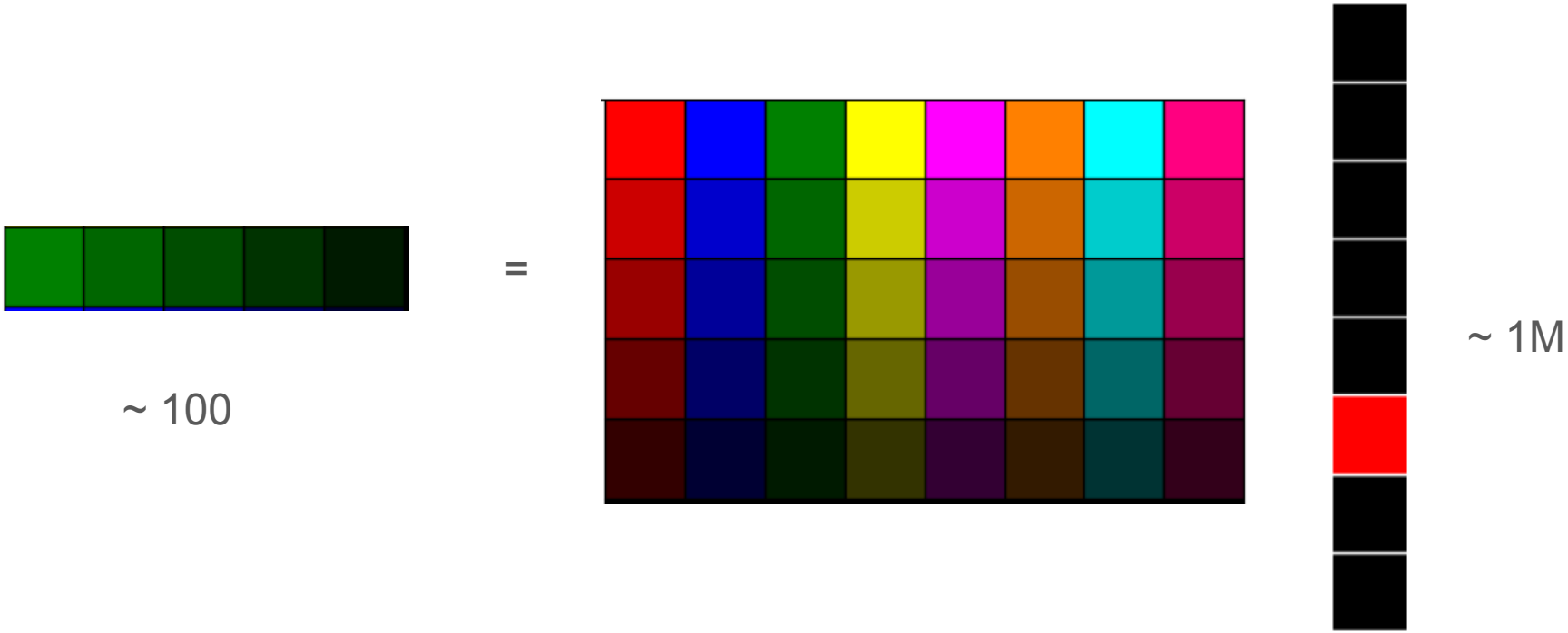
Makes features comparable

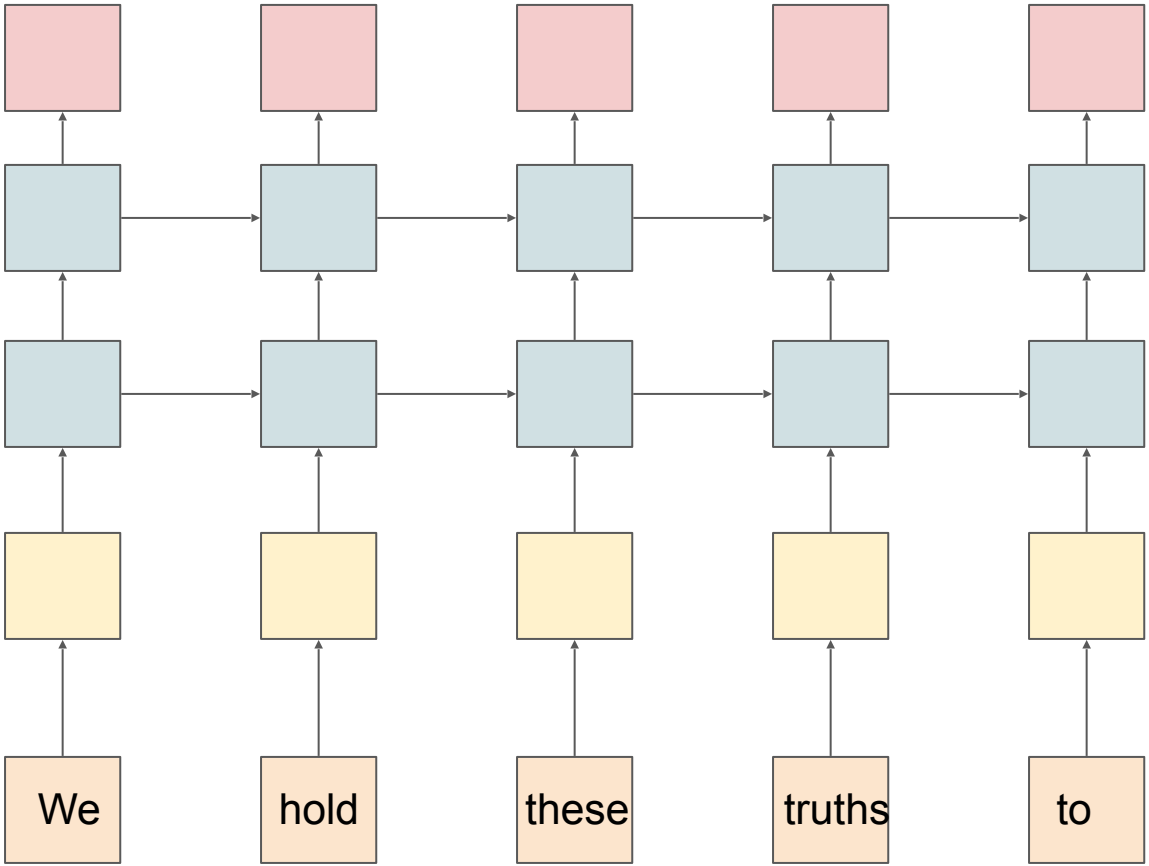
Does not encode semantics

Needs lots of data

Embedding

$$y_n = Ex_n$$




We
hold
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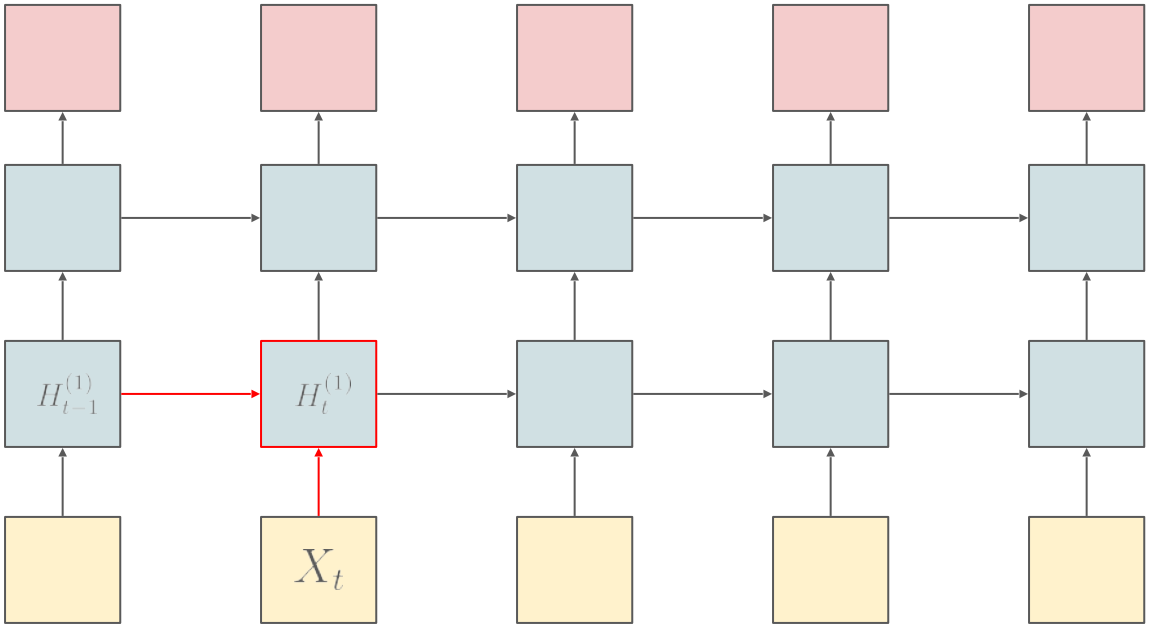
to

Stability of Recurrent Networks



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$$\ldots$$

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$$H_t = W_h^{t+1} H_{-1} + W_h^t W_x X_0 + \ldots$$

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$$W_h = \begin{bmatrix} \lambda_{max} & & \\ & \ddots & \\ & & \lambda_{min} \end{bmatrix}$$

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If largest eigenvalue of the hidden weights matrix is much bigger than 1, our hidden state will get exponentially larger, and gradients will explode.

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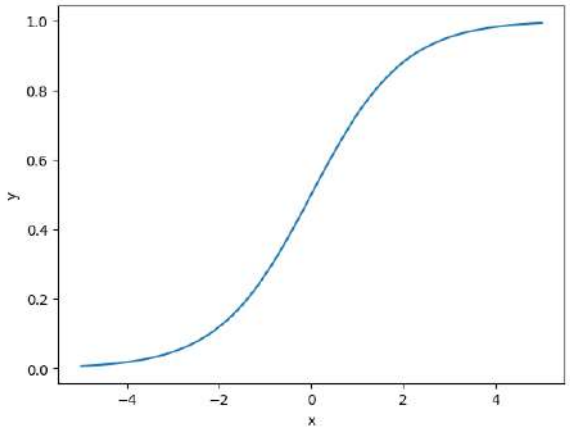
If largest eigenvalue of the hidden weights matrix is much bigger than 1, our hidden state will get exponentially larger, and gradients will explode.

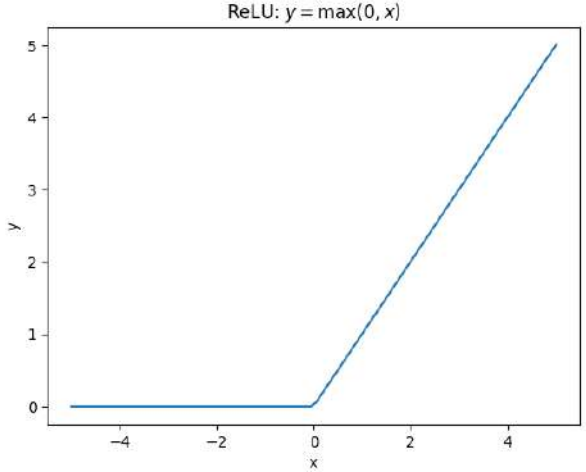
If largest eigenvalue of the hidden weights matrix is much smaller than 1, our hidden state will get exponentially small, and gradients will vanish.

What about nonlinearities? Do they help?

The sigmoid family saturates easily



The ReLu family exacerbates vanishing and exploding gradients



Recap

- Sequences
- Autoregression
- Recurrent neural networks
- Language Models
- Long Short Term Memory Networks

Next time:

- Attention
- Transformers